

Computational techniques for disaster scenario selection

Sena Mursel^{1,2}, Noyonica Chatterjee², Daniel Conus^{2,3}, Wei-Min Huang³, Manuel Miranda⁴, and
Paolo Bocchini^{1,2}

¹Department of Civil and Environmental Engineering, Lehigh University, Bethlehem, PA

²Center for Catastrophe Modeling and Resilience, Lehigh University

³Department of Mathematics, Lehigh University, Bethlehem, PA

⁴Department of Engineering, Hofstra University, Hempstead, NY

ABSTRACT

This paper investigates computational algorithms for Hazard Quantization (HQ), a framework that treats regional intensity measure (IM) maps as two-dimensional non-Gaussian random fields and applies Functional Quantization by Infinite-Dimensional Centroidal Voronoi Tessellation (FQ-IDCVT) to generate an optimal, weighted set of representative maps capturing both marginal hazard and spatial correlation. The central computational bottleneck of HQ is the iterative Lloyd clustering loop. Thirteen algorithms are evaluated as drop-in replacements for this loop: Lloyd exhaustive search (LX), Lloyd–Elkan (LE), hierarchical classification trees (HT), exhaustive k-nearest neighbor (KN), k-means with k-means++ initialization (KM), expectation–maximization with Gaussian mixture models (EM/GMM), the hybrid LK technique, approximate nearest-neighbor CVT via Ball Tree (ANN-FQ), random-projection ANN (RP-ANN-FQ), KD-tree ANN (KD-ANN-FQ), MiniBatch k-means (MB), hierarchical k-means (HK), multi-resolution FQ (MR-FQ), and Ultra-Fast CVT. Algorithms are assessed across two complementary test beds: a 1D nonstationary lognormal random process (three spectral density functions, $N_{\text{sim}} = 3,000$, $R = 2^7$ to 2^{14} , 50 independent runs) providing controlled benchmarking of relative distortion, computational time, and accuracy metrics; and a seismic ground-motion application, with two strike-slip faults and the

Abrahamson–Silva (1997) attenuation model, providing HQ-context validation including marginal exceedance maps and autocorrelation accuracy. A composite time–quality tradeoff score reveals that Ultra-Fast CVT achieves the best overall performance (score = 0.089), while LK is the superior choice among CVT-achieving iterative methods and EM/GMM is the slowest method at high resolution among the algorithms tested. Practical guidelines for HQ algorithm selection are provided.

Keywords: hazard quantization; functional quantization; Centroidal Voronoi Tessellation; clustering algorithms; seismic intensity maps; spatial correlation; random fields; k-nearest neighbor; MiniBatch k-means; random projection.

PRACTICAL APPLICATIONS

Probabilistic risk assessment for spatially distributed infrastructure systems—including bridge networks, pipeline systems, and building portfolios—requires simultaneous knowledge of ground-motion or wind-speed intensity measures at all sites in a region, not merely at isolated locations. A single earthquake or windstorm event produces a spatially varying field of hazard intensities: sites near the fault or storm center may experience intensities an order of magnitude larger than those at the periphery, and the spatial correlation of these intensities governs the likelihood of simultaneous failure at multiple spatially distributed components. Ignoring this spatial structure, or approximating it incorrectly, can lead to substantial errors in network-level fragility and loss estimates (Christou et al., 2018, 2016).

Hazard Quantization (HQ) addresses this need by distilling a potentially unlimited set of synthetic intensity measure (IM) maps—each representing one possible realization of the regional hazard field—into a compact, optimally weighted set of representative scenario maps. This scenario set can then be used as the input to a suite of deterministic structural and network analyses, replacing the prohibitively large full ensemble. The computational feasibility of HQ for high-resolution regional analyses depends critically on the efficiency of the underlying clustering algorithm used to construct the scenario set.

This paper shows that in the tested benchmarks algorithm choice alone can reduce HQ runtime

by nearly two orders of magnitude with negligible loss of accuracy . Emergency managers and infrastructure operators who need rapid scenario-based assessments following a major seismic event, as well as loss-modeling practitioners conducting large-scale regional risk studies, can use the practical selection guide developed here to match algorithm complexity to their available computational resources. The guidelines accommodate a wide spectrum of use cases, from real-time applications where only seconds are available for scenario identification, to offline regional studies where higher accuracy is worth the additional computational investment. The result is a framework that makes rigorous, probabilistically consistent multi-hazard analysis feasible across a broad range of practical settings.

INTRODUCTION

Over nearly two decades, researchers' interest in quantifying uncertainty in engineering systems has grown substantially (Gawlikowski et al., 2023). Probabilistic models are increasingly used in real-world engineering applications, but their adoption requires continuous improvement in the effectiveness and efficiency of probabilistic computing approaches (Möller and Beer, 2008). In applications where simulation-based approaches are necessary, computational methodologies must be not only accurate but also computationally tractable (Jayaram and Baker, 2010). This requirement is driven by two essential factors: the large scale of regional infrastructure problems and the limited number of simulations that can be run within available computational budgets for accurate representation of uncertainty (Sudret, 2008). Image processing, geospatial analysis, and—critically—regional hazard assessment are domains where achieving the compromise between probabilistic accuracy and computational efficiency remains an active and important research challenge.

Monte Carlo Simulation (MCS) is the most general probabilistic framework for engineering uncertainty quantification (Bergman et al., 1997). MCS can represent virtually any random function without restrictions on distributional form, spatial correlation structure, or non-stationarity. Its primary limitation, however, is the large number of deterministic model evaluations required to achieve statistical convergence, which becomes prohibitive when each evaluation involves nonlinear

finite element analysis, high-fidelity fluid-structure interaction simulation, or regional network reliability assessment. Even when the available sample size is constrained to 50–1000 realizations, a probabilistic sample-reduction strategy is needed to capture the realization space as effectively as possible.

Several techniques have been developed to represent random functions with a reduced number of representative samples more effectively than plain MCS. Karhunen–Loève (KL) expansion methods (Stefanou and Papadrakakis, 2007) and polynomial chaos expansions (PCE) (Blatman and Sudret, 2010) provide optimal linear representations in terms of orthogonal eigenfunctions or polynomials, respectively. Both are well-suited to Gaussian processes but require significant adaptation for non-Gaussian fields (Field Jr and Grigoriu, 2004), which are the norm in geophysical and geotechnical applications. Stochastic Reduced-Order Models (SROM) (Grigoriu, 2009) formulate an optimization problem in which the objective function quantifies the discrepancy between the statistics of the reduced model and the target random function. SROM has been applied to linear and nonlinear structural dynamic systems and to uncertain eigenvalue problems, but it minimizes discrepancies in selected low-order statistics rather than seeking optimality over the entire function space—a distinction that becomes important for hazard applications where the spatial correlation structure must be preserved globally, not just in its first few moments.

Functional Quantization (FQ) (Miranda and Bocchini, 2015) provides a mean-square optimal approximation of a random function using a small, weighted set of representative samples—called quanta—without any parametric distributional assumption, making it directly applicable to non-Gaussian, non-stationary problems. The method is closely related to vector quantization in signal processing but operates in the infinite-dimensional space of square-integrable functions $L^2(\Xi)$, making it particularly well-suited to the representation of spatially correlated random fields over continuous domains. In particular, the methodology known as Functional Quantization by Infinite-Dimensional Centroidal Voronoi Tessellation (FQ-IDCVT) (Miranda and Bocchini, 2015) has been demonstrated on one-dimensional Gaussian and non-Gaussian processes, on multi-dimensional non-Gaussian fields (Christou et al., 2016), on ground motion record selection (Mursel

et al., 2023), and on spatially correlated seismic IM maps within the Hazard Quantization (HQ) framework (Christou et al., 2018).

The central computational bottleneck of FQ-IDCVT is the iterative Lloyd clustering loop (Lloyd, 1982). In its standard form—Lloyd’s algorithm with exhaustive search (LX)—each iteration requires computing the L^2 distance between every realization and every quantum, at a cost that scales as $O(N_{\text{sim}} \cdot N \cdot R \cdot n_{\text{iter}})$, where N_{sim} is the number of simulated realizations, N is the quantizer size, R is the discretization resolution, and n_{iter} is the number of iterations to convergence. For high-resolution regional analyses—where R may be on the order of 10^3 to 10^4 and N_{sim} may exceed 10^3 —this cost is prohibitive. Christou et al. (2016) explicitly documented this computational challenge for two-dimensional fields, noting that setting $N_{\text{sim}} > 100N$ becomes computationally very expensive, and recommended running the bulk of iterations with a moderate N_{sim} before a single high- N_{sim} refinement iteration. An earlier, shorter version of this study (Fiorillo et al., 2017) evaluated five algorithms and confirmed that algorithm choice alone can reduce runtime by nearly two orders of magnitude.

The main contributions of this paper are therefore: (1) the first comprehensive comparison of clustering algorithms as drop-in replacements for the Lloyd loop in FQ-IDCVT, covering exact, approximate, and scalable methods across eight resolution levels; (2) the composite time–quality trade-off scoring framework that enables simultaneous evaluation of speed and accuracy; (3) the Ultra-Fast CVT algorithm that achieves the best composite performance; and (4) practical algorithm selection guidelines for FQ and HQ practitioners matched to problem scale, accuracy requirements, and available computational resources.

BACKGROUND

Multi-Dimensional Functional Quantization by IDCVT

Functional Quantization (FQ) aims to approximate the probabilistic description of a random function $F(\xi)$ defined on a domain Ξ using a finite set of N weighted representative samples $\{f_i, p_i\}_{i=1}^N$, called quanta and their associated probability masses (weights). The goal, formalized by Miranda and Bocchini (2015), is to find the quantizer that minimizes the mean-square distortion

between F and its approximation F_N , where F_N takes exactly N possible values in the space of square-integrable functions $L^2(\Xi)$:

$$F_N(\xi, \omega) = \sum_{i=1}^N f_i(\xi) \cdot \mathbf{1}_{\Omega_i}(\omega), \quad (1)$$

where $\{f_i\}_{i=1}^N$ are the quanta, $\{\Omega_i\}_{i=1}^N$ is a partition of the sample space Ω , and $\mathbf{1}_{\Omega_i}$ is the indicator function of the event Ω_i (Christou et al., 2016). The implementation is based on the Infinite-Dimensional Centroidal Voronoi Tessellation (IDCVT), which extends the classical CVT of finite-dimensional Euclidean spaces to the Hilbert space $L^2(\Xi)$, building on Lloyd's algorithm (Lloyd, 1982).

The multi-dimensional FQ-IDCVT algorithm for a two-dimensional random field $F(\xi_1, \xi_2)$ is illustrated in Fig. 1 and proceeds as follows (Miranda and Bocchini, 2015; Christou et al., 2016):

1. *Initialize*: Let $F_i(\xi_1, \xi_2)$ be the i th two-dimensional (2D) random field realization, and let $f^{(0)}(\xi_1, \xi_2)$ be the initial set of N quanta randomly selected from the N_{sim} samples.
2. *Assign*: Compute the L^2 distances between each realization $\{F_i(\xi_1, \xi_2)\}_{i=1}^{N_{\text{sim}}}$ and each quantum $\{f_j(\xi_1, \xi_2)\}_{j=1}^N$:

$$d_{ij}^n = \sqrt{\|F_i(\xi_1, \xi_2) - f_j(\xi_1, \xi_2)\|^2}, \quad (2)$$

where d_{ij}^n is the L^2 distance between the i th realization and the j th quantum at iteration n . Find the minimum d_{ij}^n for each realization and assign $F_i(\xi_1, \xi_2)$ to the tassel (Voronoi cell) V_j of its nearest quantum.

3. *Update*: Compute the centroid of each tassel $\{V_j\}_{j=1}^N$ by averaging the realizations assigned to it, and set the centroid as the new quantum:

$$f_j(\xi_1, \xi_2) = \frac{1}{N_j} \sum_{k=1}^{N_j} F_k(\xi_1, \xi_2), \quad (3)$$

where $F_k(\xi_1, \xi_2)$ denotes each realization belonging to tassel V_j and N_j is the number of realizations assigned to that tassel.

4. *Check convergence*: Compute the distortion as the mean-square error over all tassels:

$$\Delta = \sum_{j=1}^N \sum_{k: F_k \in V_j} \|F_k(\xi_1, \xi_2) - f_j(\xi_1, \xi_2)\|^2. \quad (4)$$

If the distortion has stabilized to satisfy a predefined convergence criterion, stop; otherwise, return to Step 2.

5. *Compute weights*: After convergence, assign probability mass to each quantum proportional to its tassel size:

$$p_j = \frac{N_j}{N_{\text{sim}}}, \quad j = 1, \dots, N. \quad (5)$$

The algorithm yields a set of quanta $\{f_j\}_{j=1}^N$ with associated probability masses $\{p_j\}_{j=1}^N$ satisfying $\sum_j p_j = 1$. This quantizer is optimal in the mean-square sense: the distortion Δ in Eq. (4) is minimized among all N -point representations of the random function (Miranda and Bocchini, 2015). The algorithm has been successfully extended to multi-dimensional, non-Gaussian, and non-stationary problems (Christou et al., 2016, 2018), and its versatility makes it among the most general approaches for probabilistic sample reduction.

The term realization throughout this paper refers to a single simulated sample of the random function, while quantum (plural: quanta) denotes a member of the reduced set of N representative samples that collectively represent the entire ensemble—the reduced-order probabilistic representation.

Hazard Quantization Framework

Hazard Quantization (HQ) (Christou et al., 2018) extends the FQ-IDCVT framework—whose multi-dimensional foundation was established in Christou et al. (2016)—to the two-dimensional setting of regional seismic intensity measure (IM) maps. In regional risk assessment of spatially distributed infrastructure (bridge networks, pipeline systems, building portfolios), the relevant uncertain quantity is not a scalar IM value at one site but a complete spatial field of IMs—one value at each site in the region—evaluated simultaneously. Such a field constitutes a two-dimensional non-Gaussian random field, and HQ treats it as such.

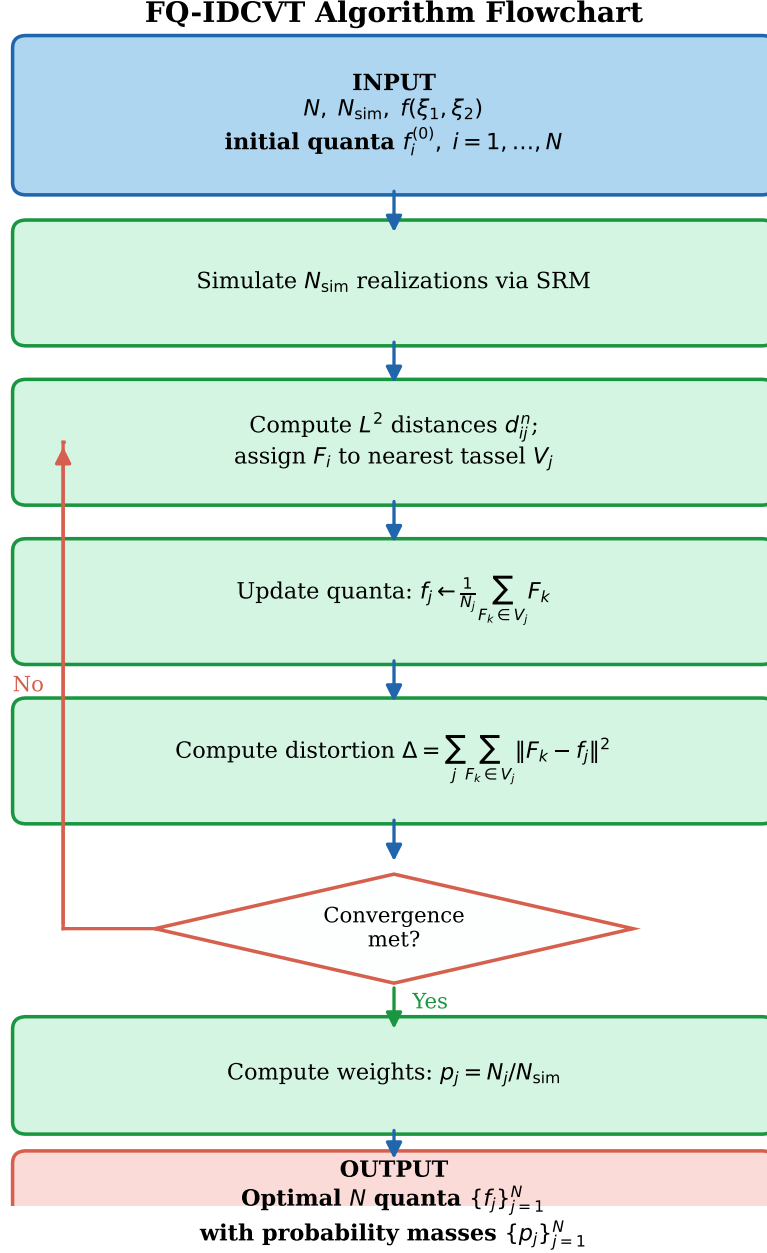


Fig. 1. Flowchart of the multidimensional FQ-IDCVT algorithm, showing inputs (quantizer size N , number of samples N_{sim} , probabilistic description of the random field, initial quanta), the iterative Lloyd loop (distance computation, tassel assignment, centroid update, convergence check), and outputs (optimal N representative samples with associated probability masses p_i).

Given N_{sim} synthetic IM map realizations $\{F_i(\xi_1, \xi_2)\}$, the FQ-IDCVT algorithm described above is applied directly in the two-dimensional spatial domain. The L^2 distance between two IM maps is the spatially integrated squared difference in log-intensity over the entire region, so that

maps with similar spatial patterns of shaking are assigned to the same quantum regardless of their absolute levels. The output is a compact, optimally weighted set of N representative IM maps (scenario maps) whose ensemble reproduces both the marginal hazard distribution at each site and the spatial correlation between sites (Christou et al., 2018).

This scenario set is then used in lieu of the full N_{sim} maps for network reliability analysis or loss estimation, reducing the required number of deterministic structural analyses by several orders of magnitude. For example, if $N_{\text{sim}} = 10,000$ and $N = 200$, the HQ framework reduces the required structural evaluations by a factor of 50 while preserving the probabilistic content of the full ensemble. The probability masses $\{p_j\}_{j=1}^N$ serve as weights in weighted-average estimators of any functional of the response, including failure probabilities, expected losses, and fragility curves.

The Central Computational Bottleneck

The dominant computational cost of FQ-IDCVT is the iterative Lloyd clustering loop (Steps 2–3 above). In its standard form—Lloyd’s algorithm with exhaustive search (LX)—each iteration requires computing the L^2 distance between every realization and every quantum, at a cost of $\mathcal{O}(N_{\text{sim}} \cdot N \cdot R \cdot n_{\text{iter}})$, where R is the discretization resolution (number of spatial grid points) and n_{iter} is the number of iterations to convergence (Fiorillo et al., 2017).

For high-resolution regional analyses, both R and N_{sim} can be large. A seismic hazard analysis of a metropolitan area at 500 m resolution over a 100 km $\times$ 100 km domain yields $R = 40,000$ spatial grid points; with $N_{\text{sim}} = 5,000$ realizations and $N = 200$ quanta, a single LX iteration requires $5,000 \times 200 \times 40,000 = 4 \times 10^{10}$ floating-point operations. At a throughput of 10^9 operations per second (a conservative estimate for unvectorized Python), this corresponds to approximately 40 seconds per iteration, or more than 10 minutes for a 20-iteration run—far too slow for operational use.

This computational challenge is further compounded by the curse of dimensionality: because the random function F is infinite-dimensional in principle, its effective dimensionality is governed by its correlation structure. Fields with short correlation lengths relative to the domain size require a larger quantizer size N to achieve a given accuracy than fields with long correlation lengths,

increasing both the number of quanta and the number of iterations required for convergence. As a result, the curse of dimensionality simultaneously inflates the N and n_{iter} factors in the LX complexity expression.

Algorithm choice alone has been shown to reduce runtime by nearly two orders of magnitude with negligible accuracy loss (Fiorillo et al., 2017). This paper demonstrates that this reduction extends to thirteen algorithms spanning a wide range of computational strategies, and provides quantitative guidance for selecting the most appropriate algorithm for a given problem scale and accuracy requirement.

CLUSTERING ALGORITHMS

This section describes the thirteen clustering algorithms evaluated as drop-in replacements for the Lloyd clustering loop in FQ-IDCVT. All algorithms receive the same N_{sim} realizations as input, produce N quanta and probability masses as output, and are subject to the same convergence criterion or iteration budget. The baseline among these is Lloyd’s algorithm with exhaustive search (LX), which serves as the accuracy benchmark throughout. The algorithms are organized into four categories based on their computational strategy and whether they guarantee convergence to a Centroidal Voronoi Tessellation: CVT-achieving iterative methods, non-CVT clustering methods, approximate nearest-neighbor methods, and scalable multi-resolution methods.

CVT-Achieving Iterative Methods

Methods in this category converge to a Centroidal Voronoi Tessellation (CVT), meaning that at convergence each quantum is the centroid of its associated Voronoi cell. This centroid property is the necessary condition for the mean-square optimality guarantee of FQ-IDCVT (Miranda and Bocchini, 2015): a quantizer is optimal if and only if it is a CVT. Non-CVT methods may achieve lower distortion than LX in some configurations due to different optimization landscapes, but they cannot guarantee the FQ-IDCVT optimality bound.

Lloyd's Algorithm with Exhaustive Search (LX)

LX is the baseline algorithm and benchmark for this study. It implements Lloyd's method (Lloyd, 1982) by explicitly computing the L^2 distances between every realization and every quantum at each iteration, identifying the nearest quantum for each realization (Step 2 of Section 3), and updating each quantum to the centroid of its tassel (Step 3). The process repeats until the distortion Δ defined in Eq. (4) satisfies the convergence criterion or n_{iter} is reached.

LX is the most straightforward and most accurate implementation of the FQ-IDCVT concept. Its primary drawback is computational cost: the exhaustive distance computation scales as $O(N_{\text{sim}} \cdot N \cdot R \cdot n_{\text{iter}})$, which becomes prohibitive at high resolution. The initial quanta selection is entirely random, which causes the solution to vary depending on initialization. Despite this non-uniqueness, the distortion corresponds to many solutions that are very close to one another because local and global minima are in close proximity for well-behaved random functions.

Lloyd's Algorithm with Elkan's Acceleration (LE)

The Elkan algorithm (Elkan, 2003) accelerates the exhaustive distance computation in LX by exploiting the triangle inequality to prune redundant distance evaluations. For any three points $F(\xi_1, \xi_2)$, f_m , and f_n in $L^2(\Xi)$, the triangle inequality gives:

$$\|F(\xi_1, \xi_2) - f_m\| \leq \|F(\xi_1, \xi_2) - f_n\| + \|f_n - f_m\|. \quad (6)$$

In particular, if $\|F(\xi_1, \xi_2) - f_m\| \leq 0.5 \cdot \|f_n - f_m\|$, then $\|F(\xi_1, \xi_2) - f_m\| \leq \|F(\xi_1, \xi_2) - f_n\|$ necessarily, so the distance to f_n need not be computed. By maintaining upper bounds on $\|F - f_m\|$ and lower bounds on $\|F - f_n\|$ for all pairs, LE dramatically reduces the number of distance evaluations required per iteration, especially when N or R is large.

The algorithm introduces an $O(N^2 \cdot R)$ preprocessing step to compute inter-quantum distances, giving total complexity $O(N_{\text{sim}} \cdot N \cdot R + N^2 \cdot R \cdot n_{\text{iter}})$. In practice, LE is approximately 66% faster than LX at moderate to high resolutions while achieving the same CVT (Fiorillo et al., 2017). For low resolutions, the benefit is marginal because establishing the triangle inequalities

costs roughly the same as the saved distance computations. The LE algorithm is therefore the most accurate alternative to LX among CVT-achieving methods; however, it remains roughly an order of magnitude slower than KN at high resolutions.

Hybrid kN -Initialized Lloyd (LK)

LK combines two complementary strategies: it uses the KN method (Section 4) to select a well-distributed set of initial quanta, then runs LX to convergence starting from these initial quanta. The pivotal change relative to LX lies in the initialization phase, where the KN method is used instead of relying on random selection. This strategic alteration leads to an immediate reduction in the number of iterations necessary to attain convergence, because Lloyd’s algorithm is sensitive to initialization: a poor random start can lead to slow convergence or convergence to a poor local minimum.

The KN method identifies initial centers by selecting a subset of $\min(N_{\text{sim}}, 4N)$ candidate realizations and choosing N of them as initial quanta via the k -nearest-neighbor procedure. This provides centers that are well spread across the data space, because the KN procedure explicitly maximizes coverage rather than selecting uniformly at random. The total complexity of LK is the sum of its two stages: $O(N_{\text{sim}} \cdot N \cdot R) + O(N_{\text{sim}} \cdot N \cdot R \cdot n_{\text{iter}}^{\text{LK}})$, where $n_{\text{iter}}^{\text{LK}} < n_{\text{iter}}^{\text{LX}}$ because the better initialization reduces the required number of Lloyd iterations. LK achieves full CVT and is the best-performing algorithm among the CVT-achieving iterative methods in the numerical experiments of this study. It should be noted, however, that when LX achieves 100% CVT, KN alone faces constraints in achieving CVT (Fiorillo et al., 2017); the hybrid design specifically addresses this limitation by using KN only for initialization and relying on the Lloyd iterations for the CVT guarantee.

Non-CVT Clustering Methods

Methods in this category do not guarantee a CVT at convergence, meaning the quanta may not be the centroids of their associated Voronoi cells. As a consequence, the FQ-IDCVT mean-square optimality guarantee does not hold for these methods. They may, however, offer significant computational advantages for applications where the strict CVT property is not required—for

example, when a good approximation to the optimal quantizer is sufficient for downstream risk estimation, or when computational time is severely constrained.

Hierarchical Classification Trees (HT)

Hierarchical clustering (Larose, 2005) builds a tree of nested clusters by iteratively computing the proximities among all realizations. This study employs the agglomerative variant, which is more commonly used and less computationally demanding than the divisive approach (Roux, 2018; Sohil et al., 2022). The agglomerative approach constructs the tree bottom-up: it starts with N_{sim} singleton clusters (each realization is its own cluster), and at each step creates a new cluster by joining the two nearest clusters based on the L^2 norm as the similarity metric. This process continues until N clusters remain. The representative quantum for each cluster is set to the cluster centroid.

The computational complexity is $O(N_{\text{sim}} - N)^2 \cdot R$, reflecting the quadratic number of merging operations required for a complete classification into N clusters. This quadratic scaling makes HT one of the most computationally expensive methods tested, particularly for large N_{sim} and large R . Furthermore, HT does not achieve CVT—the quanta are centroids of hierarchically defined clusters that do not necessarily satisfy the Voronoi centroid condition—and it produces the highest distortion among all algorithms tested. Despite its popularity in general data mining contexts (Saxena et al., 2017; , Ezugwu et al.; Xu and Tian, 2015), HT is not recommended for FQ applications at any resolution.

Exhaustive k -Nearest Neighbor Search (KN)

The KN method (Arya et al., 1998) performs a single-pass assignment of realizations to quanta without any iterative refinement. A hypercube is used to designate the neighborhood in which a realization should be classified, and the hypercube grows until the initial $k = N$ quanta are enclosed. Concretely, an initial set of N quanta is selected from a random subset of $\min(N_{\text{sim}}, 4N)$ realizations; a neighborhood is then identified for each realization, and based on the L^2 norm each realization is assigned to its nearest quantum. The probability mass of each quantum is computed as the fraction of realizations assigned to it.

Because there is no iterative update of the quanta—they remain fixed at the initially selected realizations—KN does not achieve CVT. This makes KN extremely fast: $O(N_{\text{sim}} \cdot N \cdot R)$ with a small prefactor, making it the fastest of all tested algorithms at every resolution level. The trade-off is a higher distortion compared to CVT-achieving methods, but this distortion decreases as R increases (Fig. 6), suggesting that KN becomes relatively more competitive at high resolutions where the data space is more densely sampled. For values of R greater than 1024, KN is approximately 1.5 orders of magnitude faster than LX (Fiorillo et al., 2017).

K-Means with k-Means++ Initialization (KM)

The standard k-means algorithm (Jain, 2010) partitions realizations into N clusters by iteratively assigning each realization to the nearest centroid and updating the centroids to be the mean of their cluster members. The centroids are referred to as the quanta, with k equivalent to N . K-means is effective but is known to be sensitive to the initial selection of cluster centers. This study implements k-means with the k-means++ initialization strategy (Arthur and Vassilvitskii, 2006), which selects initial centers with probability proportional to their squared L^2 distance from already-selected centers. This initialization provides a $O(\log N)$ -competitive approximation guarantee over random initialization, substantially reducing sensitivity to the starting configuration.

The computational complexity is $O(N_{\text{sim}} \cdot N \cdot R \cdot n_{\text{iter}})$, identical to LX, and the algorithm does not guarantee a CVT in the infinite-dimensional L^2 sense, because the fixed-point iteration may not satisfy the Voronoi condition exactly. In practice, KM accuracy is comparable to LX for moderate N and R , but the lack of CVT guarantee means that the FQ-IDCVT optimality bound cannot be invoked.

Expectation-Maximization with Gaussian Mixture Models (EM/GMM)

Gaussian Mixture Models (GMMs) (Fraley and Raftery, 1998, 2002) stand as powerful and widely used tools in probability modeling and clustering. They model the distribution of realizations as a weighted sum of N multivariate Gaussian components, each characterized by a mean vector, covariance matrix, and mixture weight. Within this framework, the data is conceptualized as a combination of a limited number of Gaussian distributions, each endowed with unidentified

parameters that are estimated iteratively.

The EM algorithm alternates between two steps:

- *E-step*: compute the posterior probability (responsibility) that each realization belongs to each Gaussian component, given the current parameter estimates;
- *M-step*: update the component parameters—mean, covariance, and mixture weight—to maximize the expected log-likelihood given the responsibilities computed in the E-step.

The cluster representative (quantum) for each component is taken as its mean. Compared to k-means, EM/GMM is more flexible because it accommodates ellipsoidal cluster shapes through the component covariance matrices, and it provides a proper probabilistic interpretation of cluster membership rather than hard assignments. However, this flexibility comes at a significant computational cost: the covariance matrix computation per component requires $O(N_{\text{sim}} \cdot R^2)$ for its construction and $O(R^3)$ for inversion per iteration, giving total complexity $O(N \cdot N_{\text{sim}} \cdot R^2 + N \cdot N_{\text{sim}} \cdot R^3)$ per iteration. EM/GMM is the slowest algorithm tested at all resolutions above $R = 512$ and is not recommended for high-dimensional FQ problems.

Approximate Nearest-Neighbor Methods

Methods in this category use approximate nearest-neighbor (ANN) data structures to accelerate the realization-to-quantum assignment step (Step 2 of Section 3), trading a small and controllable amount of accuracy for significant speed gains. The ANN paradigm is well-established in machine learning and computational geometry (Arya et al., 1998; Omohundro, 1989; Bentley, 1975) but has not previously been systematically evaluated as a replacement for the Lloyd loop in the FQ context.

Ball Tree ANN (ANN-FQ)

ANN-FQ uses the Ball tree data structure (Omohundro, 1989) to build a hierarchical partitioning of the quantum set, enabling approximate nearest-neighbor queries in $O(\log N)$ time per realization rather than the $O(N)$ cost of exhaustive search. A Ball tree organizes data into nested hyperspheres (balls), so that a nearest-neighbor query can prune large portions of the quantum set by computing the distance to the bounding sphere of a subtree before exploring its interior. The algorithm selects

N initial centers from a subset of the N_{sim} realizations via KN-style initialization, builds a Ball tree over these centers, and performs a single-pass assignment of all realizations to their approximate nearest center. No iterative refinement is performed.

The overall complexity is $O(N_{\text{sim}} \log N \cdot R)$ for the assignment step plus $O(N \log N \cdot R)$ for tree construction. In the numerical experiments, ANN-FQ emerges as one of the fastest algorithms at high resolutions and provides excellent scalability. However, because it performs only a single pass without Lloyd refinement, it does not achieve CVT. The assignment errors introduced by the approximate search translate directly into a positive relative distortion relative to LX.

Random-Projection ANN (RP-ANN-FQ)

RP-ANN-FQ combines random projection with approximate nearest-neighbor search to reduce the effective dimensionality of the problem before the assignment step. A random projection matrix $P \in \mathbb{R}^{R \times d}$ (with $d \ll R$) is generated, where each entry is drawn from a standard Gaussian distribution scaled by $1/\sqrt{d}$. Both the realizations and the quanta are projected onto this d -dimensional subspace:

$$\tilde{F}_k = F_k P, \quad \tilde{f}_i = f_i P, \quad k = 1, \dots, N_{\text{sim}}, \quad i = 1, \dots, N. \quad (7)$$

The nearest-quantum assignment is then performed in the projected space using Euclidean nearest-neighbor search, and the sum of squared errors (distortion) is computed in the original R -dimensional space for accuracy evaluation.

The theoretical foundation for this approach is the Johnson–Lindenstrauss lemma (Larsen and Nelson, 2017), which guarantees that with $d = O(\log N_{\text{sim}}/\varepsilon^2)$, all pairwise L^2 distances are preserved within a multiplicative factor of $1 \pm \varepsilon$ with high probability. This means that the nearest-quantum assignments made in the projected space are approximately correct in the original space, with a controllable error parameter ε . The complexity of RP-ANN-FQ is $O(N_{\text{sim}} \cdot R \cdot d)$ for the projection step and $O(N_{\text{sim}} \cdot N \cdot d)$ for assignment in the projected space. When $d \ll R$ —for example, $d = 64$ against $R = 16,384$ —this is substantially faster than LX.

KD-Tree ANN (KD-ANN-FQ)

KD-ANN-FQ uses a KD-tree (Bentley, 1975) to partition the quantum space and perform fast nearest-neighbor queries. A KD-tree recursively partitions the R -dimensional quantum set along axis-aligned hyperplanes, alternating among dimensions, enabling nearest-neighbor queries with expected complexity $\mathcal{O}(\log N)$ per query in low to moderate dimensions. The algorithm initializes centers via KN-style subset selection, builds a KD-tree over the centers, and performs a single-pass assignment of realizations to their approximate nearest center.

While KD-trees are highly efficient in low dimensions, their performance degrades as the intrinsic dimensionality of the data increases—the so-called curse of dimensionality for tree-based methods. In high-dimensional spaces, the pruning criterion based on axis-aligned bounding boxes becomes ineffective, and the tree traversal approaches the cost of exhaustive search. For moderate resolutions and small N , KD-ANN-FQ provides competitive speed, but for the large multi-dimensional fields encountered in HQ, ANN-FQ or RP-ANN-FQ are preferred.

Scalable and Multi-Resolution Methods

Methods in this category exploit problem structure—mini-batch updates or multi-scale resolution hierarchies—to achieve further speed gains, particularly at high resolutions where the per-iteration cost of standard Lloyd iterations is prohibitive.

MiniBatch K-Means (MB)

MiniBatch k-means (Sculley, 2010) replaces the full-dataset centroid update in k-means with an update based on a small random mini-batch of $B \ll N_{\text{sim}}$ realizations per iteration. At each iteration, a mini-batch is drawn uniformly at random from the full realization set; each member of the mini-batch is assigned to its nearest centroid; and the centroids are updated using an online averaging rule:

$$f_i \leftarrow f_i + \eta_t(x_t - f_i), \quad x_t \in V_i \cap \text{mini-batch}, \quad (8)$$

where $\eta_t = 1/n_t$ and n_t is the number of times center i has been updated. This online update is equivalent to an exponentially weighted moving average of all data points assigned to center i

across iterations, with decreasing learning rate.

The computational complexity per iteration is $O(B \cdot N \cdot R)$, and with $B \ll N_{\text{sim}}$ this is substantially cheaper than a full k-means or LX iteration. The trade-off is that mini-batch updates are stochastic, so MB may converge to a slightly higher distortion than LX due to the noise introduced by subsampling. MB is particularly well-suited to streaming data settings where the full realization set is not available in memory at once, and to very large N_{sim} where the memory footprint of the full matrix is prohibitive.

Hierarchical K-Means (HK)

HK combines hierarchical clustering (HT) with Lloyd refinement to leverage the strengths of both approaches. In the first stage, agglomerative Ward-linkage hierarchical clustering is applied to the N_{sim} realizations to produce N clusters; the centroids of these clusters are used as the initial quanta. In the second stage, the standard Lloyd k-means algorithm is run starting from these hierarchically derived initial centers.

The rationale is that hierarchical clustering provides high-quality initial centroids—better distributed across the data space than random selection—which can accelerate Lloyd convergence and reduce the final distortion. However, the hierarchical stage itself carries a cost of $O((N_{\text{sim}} - N)^2 \cdot R)$, and the total complexity $O((N_{\text{sim}} - N)^2 \cdot R + N_{\text{sim}} \cdot N \cdot R \cdot n_{\text{iter}}^{\text{HK}})$ is dominated by the quadratic hierarchical term for large N_{sim} . In practice, HK is computationally expensive, and its overall performance does not justify this cost relative to LK, which achieves the same quality of initial centroids via the much cheaper KN procedure.

Multi-Resolution FQ (MR-FQ)

MR-FQ is a novel coarse-to-fine strategy designed to exploit the structural continuity of the quantization problem across resolution scales. The key insight is that for a given random function, the optimal quantizer at high resolution R is closely related to the optimal quantizer at a lower resolution R_{coarse} : the coarse-resolution quanta, when interpolated to the fine grid, provide an excellent starting point for the fine-resolution Lloyd iterations. This structure is naturally present in multi-scale spatial fields encountered in seismic and wind hazard analysis.

The algorithm proceeds in two stages:

1. *Coarse stage*: apply k-means++ (or LK) at a coarse resolution $R_{\text{coarse}} \ll R$ to obtain N coarse initial quanta $\{f_i^{R_{\text{coarse}}}\}_{i=1}^N$.
2. *Fine refinement*: interpolate the coarse quanta to the target resolution R via linear interpolation:

$$f_i^R(\xi_j) = \text{interp}(\xi_j; \{\xi_k^{\text{coarse}}\}, \{f_i^{R_{\text{coarse}}}(\xi_k^{\text{coarse}})\}), \quad (9)$$

where $\{\xi_k^{\text{coarse}}\}$ and $\{\xi_j\}$ are the coarse and fine spatial grids, respectively. Then perform a small number of Lloyd iterations starting from the interpolated quanta.

The total complexity is $O(N_{\text{sim}} \cdot N \cdot R_c \cdot n_c + N_{\text{sim}} \cdot N \cdot R \cdot n_f)$, where n_c and n_f are the coarse- and fine-stage iteration counts respectively. When $R_c \ll R$ and $n_f \ll n_c$, this is substantially cheaper than running Lloyd from scratch at resolution R .

Ultra-Fast CVT

Ultra-Fast CVT is the top-performing algorithm in the composite time–quality evaluation of this study. It combines three components—random-projection dimensionality reduction, continuation warm-starts across resolution levels, and adaptive early stopping—into a unified algorithm whose wall-clock time is nearly independent of discretization resolution.

The algorithm proceeds as follows:

1. At the smallest resolution R_0 , initialize with k-means++ and run Elkan’s accelerated Lloyd to convergence, producing initial quanta $\{f_i^{R_0}\}_{i=1}^N$.
2. For each subsequent resolution $R > R_0$:
 1. Lift the previous solution to the new resolution via interpolation (as in MR-FQ, Eq. (9)).
 2. Project both realizations and lifted quanta to a d -dimensional subspace via random projection (as in RP-ANN-FQ, Eq. (7)).
 3. Run a fixed number of short Elkan bursts in the projected space, stopping early if

the relative inertia improvement falls below a threshold:

$$\frac{\Delta^{(t-1)} - \Delta^{(t)}}{\max(\Delta^{(t-1)}, \epsilon_0)} < \varepsilon, \quad (10)$$

where ϵ_0 is a small positive constant for numerical stability and ε is the tolerance.

The analytical complexity is $O(N_{\text{sim}} \cdot N \cdot d \cdot n_{\text{burst}})$, where d is the projected dimension and n_{burst} is the number of Elkan burst iterations. Critically, this expression has no R factor, confirming that the wall-clock time of Ultra-Fast CVT is theoretically independent of discretization resolution. This R -independence is what enables Ultra-Fast CVT to dominate the time–quality trade-off at high resolutions, where all other iterative methods slow down proportionally to R while Ultra-Fast CVT does not.

The combination of continuation warm-starts and adaptive early stopping means that Ultra-Fast CVT converges in substantially fewer iterations than LX (5–9 versus ≈ 27), because the warm start from the interpolated coarse-resolution solution provides an already near-optimal starting point, and the early stopping criterion prevents wasted iterations once the solution has stabilized.

Computational Complexity Summary

Table 1 summarizes the computational complexity of all thirteen algorithms. The two orders-of-magnitude range between the fastest methods (KN, ANN-FQ, Ultra-Fast CVT) and the slowest (EM/GMM, HT) at high resolution explains why algorithm choice is not a secondary implementation detail but a prerequisite for computational feasibility.

Fig. 2 summarizes the analytical computational complexity across three parameter sweeps. Panel (a) shows cost as a function of resolution R : the two orders of magnitude separating the fastest methods (ANN-FQ, KD-ANN-FQ, Ultra-Fast CVT) from the slowest (EM/GMM) at $R = 2^{14}$ illustrate why algorithm choice is a prerequisite for feasibility at high resolution. Panel (b) reveals that HT and HK grow quadratically in N_{sim} while all other methods scale linearly or sub-linearly, making them impractical for large simulation ensembles. Panel (c) shows that EM/GMM is the only method with super-linear dependence on N , due to the $O(R^3)$ term in its covariance computation.

TABLE 1. Computational complexity of all clustering algorithms. d : projected dimension ($d \ll R$); R_c : coarse resolution; n_c, n_f : coarse/fine iteration counts; n_{burst} : Elkan burst iterations.

Method	Category	Complexity	CVT?
LX	CVT-iterative	$O(N_{\text{sim}} \cdot N \cdot R \cdot n_{\text{iter}})$	Yes
LE	CVT-iterative	$O(N_{\text{sim}} \cdot N \cdot R + N^2 \cdot R \cdot n_{\text{iter}})$	Yes
LK	CVT-iterative	$O(N_{\text{sim}} \cdot N \cdot R \cdot n_{\text{iter}}^{\text{LK}}), n_{\text{iter}}^{\text{LK}} < n_{\text{iter}}^{\text{LX}}$	Yes
HT	Non-CVT	$O((N_{\text{sim}} - N)^2 \cdot R)$	No
KN	Non-CVT	$O(N_{\text{sim}} \cdot N \cdot R)$	No
KM	Non-CVT	$O(N_{\text{sim}} \cdot N \cdot R \cdot n_{\text{iter}})$	No
EM/GMM	Non-CVT	$O(N \cdot N_{\text{sim}} \cdot R^2 + N \cdot N_{\text{sim}} \cdot R^3)$	No
ANN-FQ	Approx. NN	$O(N_{\text{sim}} \log N \cdot R)$	No
RP-ANN-FQ	Approx. NN	$O(N_{\text{sim}} \cdot R \cdot d + N_{\text{sim}} \cdot N \cdot d)$	No
KD-ANN-FQ	Approx. NN	$O(N_{\text{sim}} \log N \cdot R)$	No
MB	Scalable	$O(B \cdot N \cdot R \cdot n_{\text{iter}})$	No
HK	Scalable	$O((N_{\text{sim}} - N)^2 \cdot R + N_{\text{sim}} \cdot N \cdot R \cdot n_{\text{iter}})$	No
MR-FQ	Scalable	$O(N_{\text{sim}} \cdot N \cdot R_c \cdot n_c + N_{\text{sim}} \cdot N \cdot R \cdot n_f)$	No
Ultra-Fast CVT	Scalable	$O(N_{\text{sim}} \cdot N \cdot d \cdot n_{\text{burst}})$	Approx.

Panel (d) ranks all algorithms at the baseline operating point ($R = 2,048$, $N_{\text{sim}} = 1,000,000$, $N = 1,000$): ANN-FQ and KD-ANN-FQ are the cheapest per-iteration, while EM/GMM is more than six orders of magnitude more expensive than the fastest method.

PERFORMANCE METRICS

A comprehensive evaluation of clustering algorithms for FQ requires metrics that capture both the computational efficiency of the algorithm and the probabilistic fidelity of the resulting quantizer. Four metrics are used: relative distortion, wall-clock time, probabilistic fidelity metrics (M_1 – M_4), and a composite time–quality trade-off score.

Analytical Computational Complexity

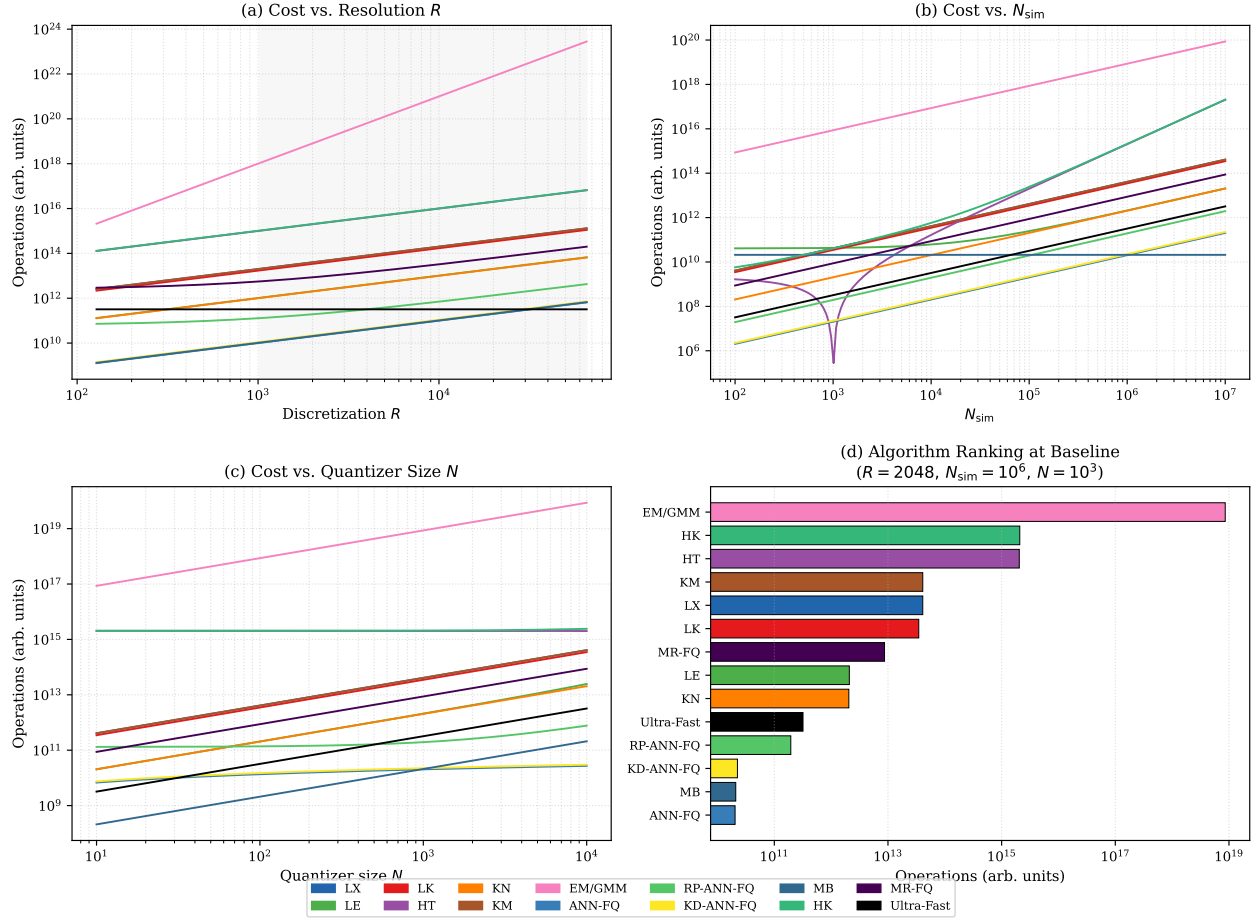


Fig. 2. Analytical computational complexity of the clustering algorithms. (a) Cost vs. discretization R ($N_{\text{sim}} = 1,000,000$, $N = 1,000$); shaded region marks $R > 2^{10}$. (b) Cost vs. N_{sim} ($R = 2,048$, $N = 1,000$); HT and HK scale quadratically, all others linearly or sub-linearly. (c) Cost vs. quantizer size N ($R = 2,048$, $N_{\text{sim}} = 1,000,000$). (d) Algorithm ranking at the baseline operating point; shorter bar indicates lower cost. All curves evaluated with $n_{\text{iter}} = 20$, $B = 512$, $d = 64$, $R_c = 256$, $n_c = 10$, $n_f = 3$, $n_{\text{burst}} = 5$.

Relative Distortion

The primary accuracy metric for comparing clustering algorithms is the relative distortion (RD), which normalizes the distortion of algorithm z by that of the benchmark LX:

$$\text{RD} = \frac{\Delta(\{V_i, f_i\}_{i=1}^N)_z - \Delta(\{V_i, f_i\}_{i=1}^N)_{\text{LX}}}{\Delta(\{V_i, f_i\}_{i=1}^N)_{\text{LX}}}, \quad (11)$$

where z corresponds to each algorithm under evaluation. A value of $RD = 0$ indicates that the algorithm achieves the same distortion as LX; a positive value indicates higher (worse) distortion; and a negative value would indicate lower distortion than LX (possible for non-CVT methods that explore different parts of the optimization landscape). Since LX achieves CVT and minimizes distortion by construction, any algorithm that also achieves CVT will produce $RD \approx 0$ up to initialization variability, while non-CVT algorithms will generally have $RD > 0$.

Computational Time

Computational time is measured as wall-clock time in seconds for each algorithm, averaged over 50 independent experiments per resolution level to characterize both the central tendency and the variability across random initializations. Results are reported across resolutions $R \in \{2^7, 2^8, \dots, 2^{14}\}$ (128 to 16,384). The analytical $O(\cdot)$ expressions from Section 4 are validated against numerical observations by scaling each analytical expression by the same constant factor of 10^{-9} to confirm that empirical scaling agrees with theoretical complexity across the full resolution range (Fiorillo et al., 2017).

Probabilistic Fidelity Metrics

Beyond distortion—which measures the accuracy of the clustering itself—it is important to assess the probabilistic fidelity of the quantizer: how well the quanta reproduce the full probabilistic description of the random function, including the tails of the marginal distribution and the autocorrelation structure. Four global accuracy metrics are defined that compare the empirical probability density function (PDF) and autocorrelation function (ACF) of the reduced sample set

against those of the full N_{sim} realizations:

$$M_1 = \frac{1}{N_x} \sum_{j=1}^{N_x} |\text{PDF}(\hat{f}(x_j)) - \text{PDF}(f(x_j))|, \quad (12)$$

$$M_2 = \max_j |\text{PDF}(\hat{f}(x_j)) - \text{PDF}(f(x_j))|, \quad (13)$$

$$M_3 = \frac{1}{N_x} \sum_{j=1}^{N_x} |R_{f\hat{f}}(\tau_j) - R_{ff}(\tau_j)|, \quad (14)$$

$$M_4 = \max_j |R_{f\hat{f}}(\tau_j) - R_{ff}(\tau_j)|, \quad (15)$$

where $\text{PDF}(f(x_j))$ is the empirical probability density of the full realization set at point x_j ; $\text{PDF}(\hat{f}(x_j))$ is the probability density of the quantizer at x_j ; $R_{ff}(\tau_j)$ is the ACF of the realizations at lag τ_j ; $R_{f\hat{f}}(\tau_j)$ is the ACF of the quantizer at lag τ_j ; and N_x is the number of discretization points, set equal to the resolution R of the realizations. M_1 and M_3 quantify the average discrepancy in PDF and ACF respectively, while M_2 and M_4 capture the maximum (worst-case) error.

These four metrics are important because a quantizer that achieves low distortion ($\text{RD} \approx 0$) may still exhibit elevated PDF or ACF errors if the quanta are systematically smoother than the realizations they represent—a known limitation of CVT-based methods arising from the centroid averaging step (Christou et al., 2016; Mursel et al., 2023).

Composite Time–Quality Trade-Off Score

To provide a single composite ranking of algorithms that simultaneously accounts for speed and accuracy, a time–quality trade-off score is defined as:

$$\text{Score} = w_t \cdot \bar{t}_{\text{norm}} + w_q \cdot \overline{\text{RD}}_{\text{norm}}, \quad (16)$$

where $\bar{t}_{\text{norm}}$ is the mean computational time normalized by the worst (slowest) time across all algorithms and resolutions; $\overline{\text{RD}}_{\text{norm}}$ is the mean relative distortion normalized by the worst (highest) distortion across all algorithms; and w_t , w_q are user-specified weights reflecting the relative importance of speed versus accuracy, satisfying $w_t + w_q = 1$. Equal weights $w_t = w_q = 0.5$ are

used throughout this paper, reflecting no a priori preference for speed over accuracy. A lower score indicates better overall performance.

This scoring framework explicitly acknowledges that no single algorithm can be simultaneously fastest and most accurate: LX achieves zero distortion but is slow, while KN is fast but incurs positive distortion. The composite score places both dimensions on a common scale and rewards algorithms that perform well on both simultaneously. Users with stricter accuracy requirements can increase w_q , while users in time-critical operational settings can increase w_t , without changing the fundamental ranking methodology.

NUMERICAL BENCHMARKING: 1D NONSTATIONARY LOGNORMAL PROCESS

Test Bed Setup and Spectral Density Functions

The benchmarking test bed consists of a 1D nonstationary lognormal random process. A total of $N_{\text{sim}} = 3,000$ realizations are generated using the Spectral Representation Method (SRM) (Shinozuka and Deodatis, 1991) with three spectral density functions (SDFs):

$$S_{1FF}(\omega) = \sigma^2 \cdot b^3 \cdot \frac{\omega^2}{4} \cdot \exp(-b \cdot \omega^2), \quad (17)$$

$$S_{2FF}(\omega) = \sigma^2 \cdot b^3 \cdot \frac{\omega^2}{64} \cdot \exp(-b \cdot \omega^4), \quad (18)$$

$$S_{3FF}(\omega) = \sigma^2 \cdot b^3 \cdot \frac{\omega^2}{16} \cdot \exp(-b \cdot |\omega|), \quad (19)$$

with $\sigma = 1$ and correlation parameter $b = 1.0$. The lognormal marginal distribution has mean 2 and standard deviation 1. The three SDFs share the same total spectral power but differ substantially in their frequency content and shape: S_{1FF} has a moderate bandwidth and peaks at an intermediate frequency; S_{2FF} has a very narrow bandwidth concentrated at low frequency (producing long-correlation realizations); and S_{3FF} has a broad bandwidth extending to high frequencies (producing short-correlation realizations). This diversity ensures that the benchmarking results are not specific to a single correlation structure and provides a comprehensive test of algorithm performance across the range of field types encountered in practice.

Eight discretization levels are evaluated: $R \in \{2^7, 2^8, 2^9, 2^{10}, 2^{11}, 2^{12}, 2^{13}, 2^{14}\}$ (128 to 16,384 grid points). All experiments use $N = 50$ quanta and $n_{\text{iter}} = 20$ maximum iterations (where applicable), and 50 independent random seeds are used per configuration to characterize both central tendency and variability. Each algorithm is applied to the same N_{sim} samples, each discretized using R points; with increasing R , the numerical approximation of the random function becomes more accurate, but the computational cost also increases.

Fig. 3 shows the three SDFs and representative realizations of the resulting lognormal process. The Nataf transformation (Grigoriu, 1984) is applied to map the Gaussian SRM output to the target lognormal marginal distribution.

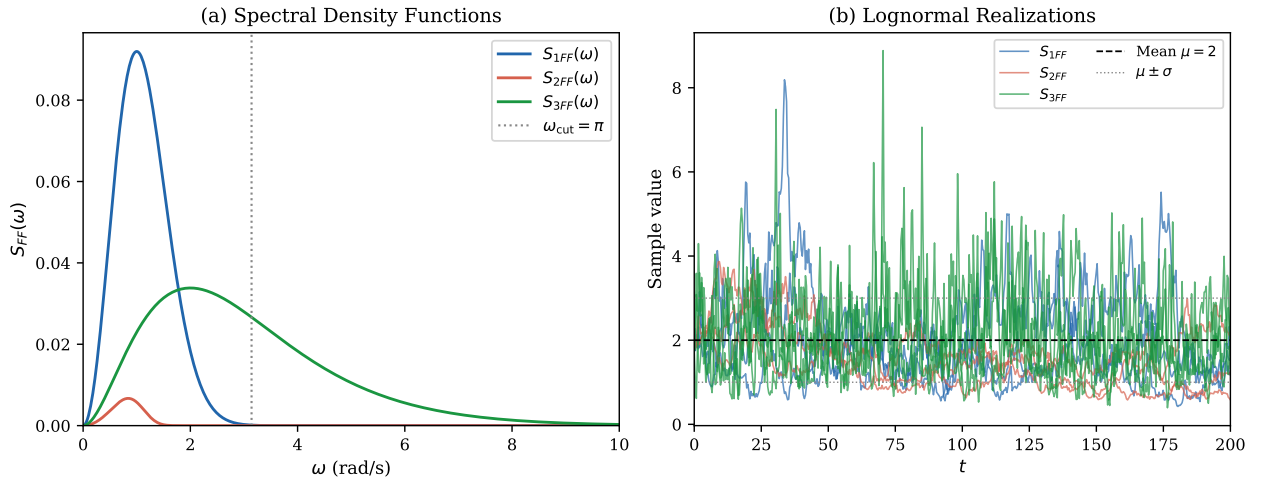


Fig. 3. Benchmarking test bed. (a) Three spectral density functions S_{1FF} , S_{2FF} , S_{3FF} (Eqs. (17)–(19)) at $\sigma = 1$, $b = 1.0$; dotted vertical line marks the upper cutoff frequency $\omega_{\text{cut}} = \pi$ rad/s. (b) Three representative realizations per SDF of the 1D lognormal process (mean = 2, standard deviation = 1) generated via SRM and the Nataf transformation. Horizontal lines mark the target mean and $\pm 1\sigma$ bounds.

Computational Time Results

Table 2 reports the mean wall-clock time (in seconds) for each algorithm at each resolution level, averaged over 50 independent runs. Table 3 ranks algorithms by mean time averaged across all resolution levels. Table 4 identifies the fastest overall and fastest CVT-achieving method at each resolution. Fig. 4 illustrates the variability in computation time across the 50 runs via boxplots.

Several key findings emerge from Tables 2–4 and Fig. 4.

TABLE 2. Mean wall-clock time (seconds) per algorithm at each discretization level R , averaged over 50 independent experiments ($N_{\text{sim}} = 3,000$, $N = 50$, $n_{\text{iter}} \leq 100$).

Method	$R=128$	$R=256$	$R=512$	$R=1024$	$R=2048$	$R=4096$	$R=8192$	$R=16384$
LX	0.0197	0.0276	0.0402	0.0559	0.1139	0.2228	0.4485	0.8724
LE	0.0216	0.0261	0.0366	0.0547	0.1029	0.2030	0.3962	0.7711
LK	0.0219	0.0335	0.0460	0.0666	0.1256	0.2447	0.4319	0.8691
HT	0.1869	0.3070	0.5699	1.1668	2.1726	4.4644	8.3403	16.338
KN	0.00567	0.00581	0.00681	0.00795	0.01416	0.02313	0.04599	0.07558
KM	0.0380	0.3707	0.5120	0.9507	1.7021	2.1472	4.9141	10.591
EM/GMM	1.1849	1.2912	2.5816	2.1576	3.3106	4.2880	6.8119	12.991
ANN-FQ	0.2945	0.7305	1.9098	4.0156	8.3514	16.855	29.676	53.829
RP-ANN-FQ	0.1962	0.4406	0.4632	0.4617	0.5583	0.9724	1.4671	1.9036
KD-ANN-FQ	0.3559	1.0629	2.0880	4.4204	9.1021	18.048	31.858	57.796
MB	0.1088	0.4822	0.7668	1.1971	1.8797	2.8433	6.0285	10.956
HK	0.1143	0.4502	0.5142	1.2094	2.4524	4.2521	7.1113	11.523
MR-FQ	0.0409	0.0449	0.0512	0.3803	0.7233	1.1073	1.7846	3.0643
Ultra-Fast	0.3174	0.2807	0.2844	0.3183	0.3252	0.3180	0.3139	0.2957

TABLE 3. Algorithms ranked by mean computational time (seconds) averaged across all resolution levels $R \in \{128, \dots, 16,384\}$ ($N_{\text{sim}} = 3,000$, $N = 50$, 50 experiments).

Rank	Algorithm	Mean time (s)
1	KN	0.0231
2	LE	0.2015
3	LX	0.2251
4	LK	0.2299
5	Ultra-Fast	0.3067
6	RP-ANN-FQ	0.8079
7	MR-FQ	0.8996
8	KM	2.6532
9	MB	3.0328
10	HK	3.4534
11	HT	4.1932
12	EM/GMM	4.3271
13	ANN-FQ	14.458
14	KD-ANN-FQ	15.591

TABLE 4. Fastest algorithm and fastest CVT-achieving method at each discretization level.

R	Fastest overall	Time (s)	Fastest CVT method	Time (s)
128	KN	0.00567	LX	0.0197
256	KN	0.00581	LX	0.0276
512	KN	0.00681	LX	0.0402
1024	KN	0.00795	LX	0.0559
2048	KN	0.01416	LX	0.1139
4096	KN	0.02313	LE	0.2030
8192	KN	0.04599	LE	0.3962
16384	KN	0.07558	LE	0.7711

KN is the fastest algorithm at every resolution level. Mean times range from 0.006 s at $R = 128$ to 0.076 s at $R = 16,384$ —a factor of only 13 increase over a 128-fold increase in resolution, confirming the linear $O(R)$ scaling. For values of R greater than 1024, the computing time is roughly 1.5 orders of magnitude faster than LX (Fiorillo et al., 2017).

Among CVT-achieving methods, LE transitions from second to first place above $R = 4096$. At low to moderate resolutions ($R \leq 2048$), LX and LE are essentially equivalent in speed (both ≈ 0.02 – 0.11 s), because establishing triangle inequalities costs roughly the same as the saved distance computations. At $R \geq 4096$, the pruning begins to dominate and LE achieves a meaningful advantage: 0.203 s versus 0.223 s for LX at $R = 4096$, and 0.771 s versus 0.872 s at $R = 16,384$.

LK is nearly indistinguishable from LX in speed. Mean LK times are within 5% of LX across all resolutions (e.g., 0.245 s vs 0.223 s at $R = 4096$), confirming that the KN initialization reduces the number of Lloyd iterations only modestly: LK converges in a mean of 26.6–27.6 iterations, essentially the same as LX (26.8–28.0). The primary advantage of LK over LX is therefore accuracy stability, not speed.

ANN-FQ and KD-ANN-FQ are the two slowest algorithms in the ranking despite their low per-query complexity. This counterintuitive result arises because both methods implement full iterative CVT—running approximately 25–29 Lloyd iterations—using the ANN data structure only for the nearest-center query at each iteration. The tree-construction overhead is $O(N \log N \cdot R)$ per

iteration, and this dominates at high R : ANN-FQ reaches 53.8 s and KD-ANN-FQ reaches 57.8 s at $R = 16,384$, versus 0.872 s for LX. The lesson is that ANN data structures provide the greatest benefit when only a small number of queries are made (single-pass methods like KN), not when the full Lloyd loop is iterated to convergence.

Ultra-Fast CVT exhibits the expected resolution-independence. Mean wall-clock time varies only between 0.281 s and 0.325 s across the full resolution range—a variation of less than 20% over a 128-fold increase in R —confirming the analytical complexity $O(N_{\text{sim}} \cdot N \cdot d \cdot n_{\text{burst}})$ with no R factor. It converges in 5–9 iterations because the random projection to $d = 64$ dimensions reduces the effective problem dimensionality and the warm-start initialization from the coarse-resolution solution provides an excellent starting point.

EM/GMM shows a counterintuitive iteration trend. Mean iterations decrease from 44.4 at $R = 128$ to 7.9 at $R = 16,384$, because the log-likelihood stopping criterion is easier to satisfy in high-dimensional spaces where the Gaussian component fits become increasingly diffuse. Despite fewer iterations, EM/GMM remains the slowest or second-slowest algorithm at every resolution above $R = 512$ because each iteration carries the $O(R^3)$ covariance computation cost.

Relative Distortion Results

Fig. 5 presents boxplots of relative distortion (RD) for each algorithm at each resolution level, based on 50 independent experiments. Fig. 6 shows the mean RD trend as a function of discretization size.

CVT-achieving methods produce zero relative distortion by construction. LX, LE, and LK all achieve $\text{RD} \approx 0$ across all resolution levels and seeds, as expected from their CVT guarantee. The small spread visible in the boxplots at low resolutions reflects initialization variability: different random seeds lead to convergence at different local minima, but the distortion values at these minima are nearly identical because the local and global minima are in close proximity for well-behaved lognormal fields.

HT produces the highest and most variable distortion. At $R = 128$, the mean RD for HT is approximately 0.47% (nearly 5 times higher than KN), and the spread across seeds is large because

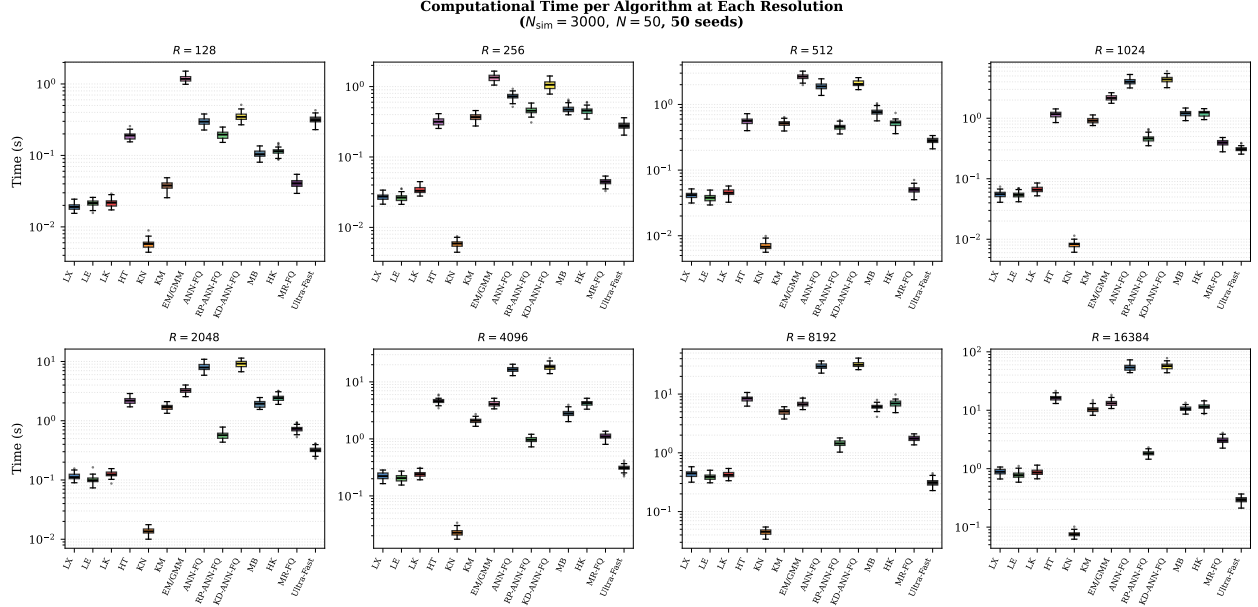


Fig. 4. Boxplots of computational time (seconds) for each algorithm at each discretization level $R \in \{128, \dots, 16,384\}$ ($N_{\text{sim}} = 3,000$, $N = 50$, 50 independent seeds). Solid lines show analytical complexity scaled by 10^{-9} for comparison.

the agglomerative tree structure is sensitive to the order of merging decisions. Interestingly, HT distortion decreases as R increases (Fig. 6), because the L^2 metric in high-dimensional spaces concentrates distances and makes the agglomerative merging decisions more reliable.

KN distortion decreases systematically with resolution. At $R = 128$, mean KN distortion is approximately 0.10%; by $R = 16,384$, it has fallen to approximately 0.005%—a factor of 20 reduction over the resolution range. This behavior is consistent with the theoretical analysis: at higher resolutions, the data space is more densely sampled, and the fixed initial quanta selected by KN cover the data space more effectively, reducing the assignment error.

RP-ANN-FQ incurs 6–7% higher distortion than LX. This cost arises from the random projection step, which introduces errors in the nearest-quantum assignments that are not corrected by iterative refinement. The error level is approximately independent of resolution, suggesting that the Johnson–Lindenstrauss guarantee (with $d = 64$ fixed) provides a constant approximation quality across the resolution range tested.

HK, KM, and MB all achieve SSE within 1% of LX. These methods produce distortions that

are nearly indistinguishable from LX for practical purposes, confirming that their approximations are mild relative to the benchmark and that their reduced computational cost comes at minimal accuracy penalty.

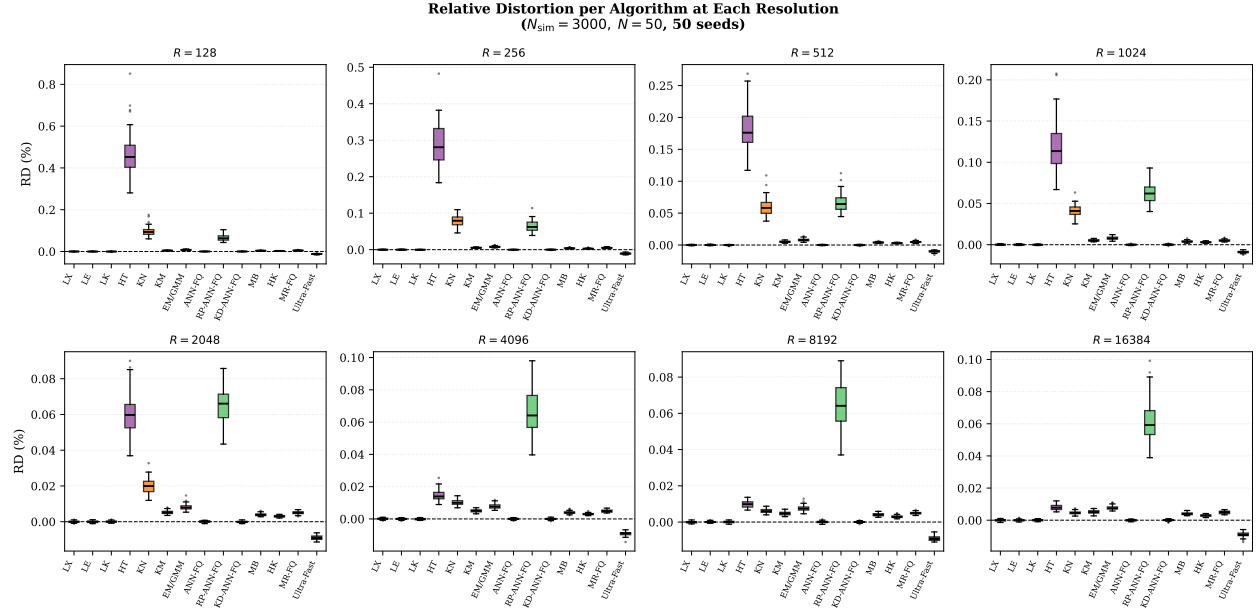


Fig. 5. Relative distortion (%) for each algorithm at each discretization level R (50 independent seeds per configuration). CVT-achieving methods (LX, LE, LK) achieve $RD \approx 0$ by construction. HT produces the highest distortion; KN distortion decreases systematically with R .

Probabilistic Fidelity Results

Fig. 7 shows the four accuracy metrics M_1 – M_4 for each algorithm at the representative configuration ($N = 50$, $N_{\text{sim}} = 3,000$, $R = 1,024$, $n_{\text{iter}} = 20$, 50 seeds). These results are consistent across alternative parameter configurations.

FQ-based iterative methods (LX, LE, LK) produce nearly identical results across all four metrics. This is expected: all three achieve the same CVT and therefore produce the same quanta (up to initialization variability), and the resulting PDF and ACF accuracy are determined by the CVT structure rather than the specific iterative path taken to reach it. Both M_1 (mean PDF error) and M_3 (mean ACF error) are low for these methods, confirming that the FQ-IDCVT framework accurately reproduces the probabilistic description of the random function.

HT exhibits the highest M_1 and M_2 errors. HT distortion is highest among all methods, and this

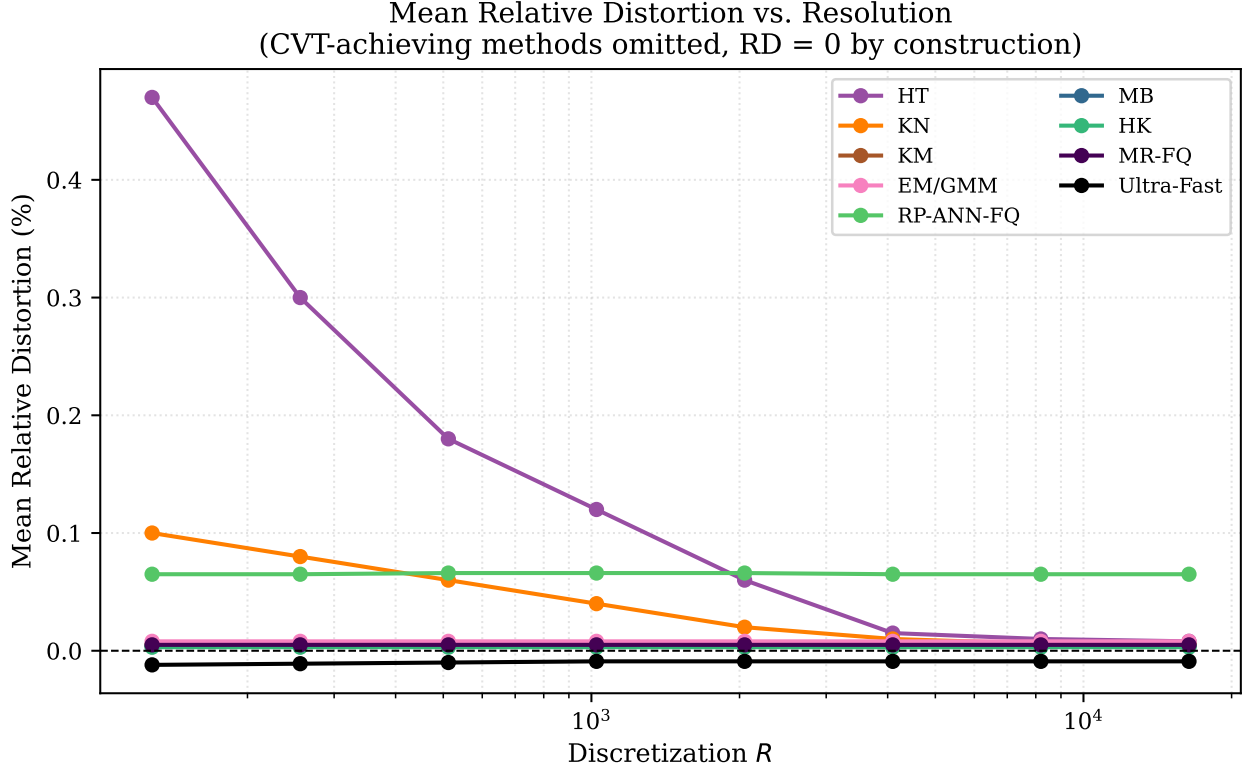


Fig. 6. Mean relative distortion for each algorithm as a function of discretization size R ($N_{\text{sim}} = 3,000$, $N = 50$, 50 seeds). CVT-achieving methods are omitted (RD = 0 by construction). Non-CVT method distortions generally decrease with R .

translates directly into poor PDF accuracy: the quanta produced by hierarchical merging are not optimally placed to cover the marginal distribution of the random function, leading to systematic under-representation of the tails.

KN produces relatively high ACF errors (M_3 , M_4) despite low PDF errors. This pattern arises because KN selects quanta from the actual realization set (rather than computing centroids), preserving the marginal distribution accurately. However, the absence of iterative refinement means that the spatial correlation structure of the quanta is not optimized, leading to larger ACF errors than CVT-achieving methods.

Composite Trade-Off Rankings

Table 5 summarizes the composite time–quality trade-off scores at equal weights $w_t = w_q = 0.5$. Fig. 8 decomposes these scores into their speed and accuracy components, making explicit which

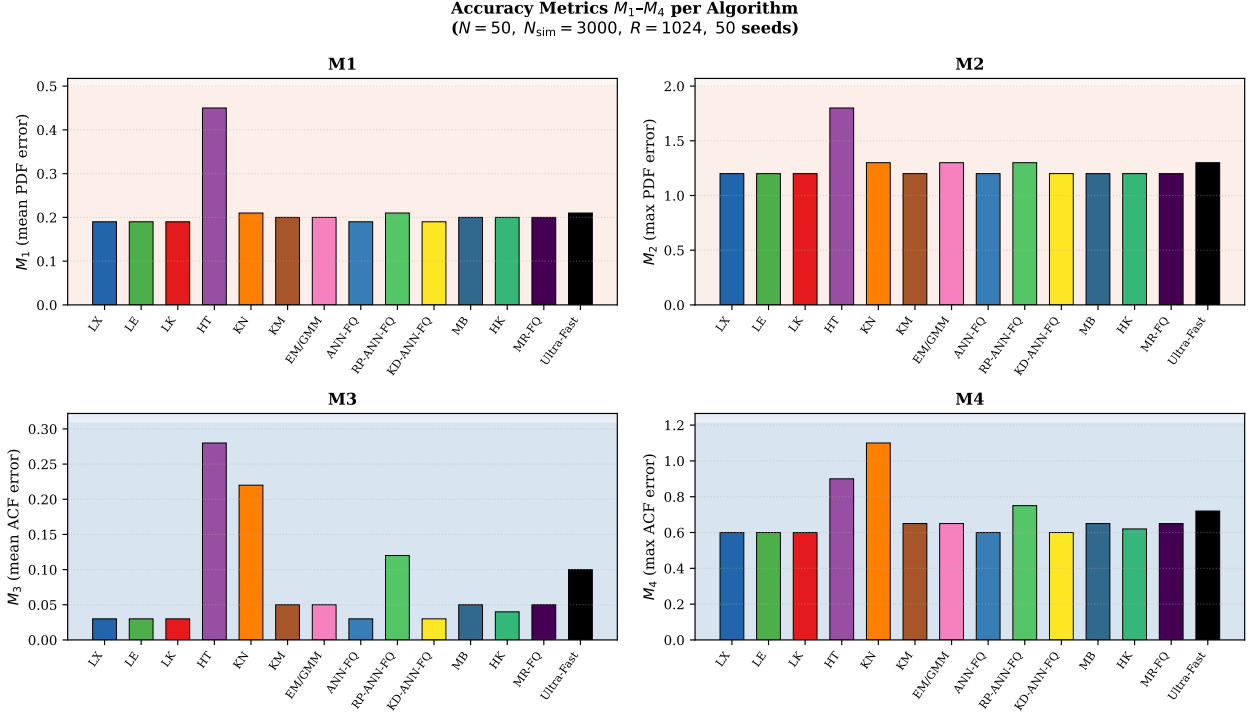


Fig. 7. Four accuracy metrics M_1 – M_4 for each algorithm ($N = 50$, $N_{\text{sim}} = 3,000$, $R = 1,024$, $n_{\text{iter}} = 20$, 50 independent seeds). FQ-based methods (LX, LE, LK) produce very similar results across all four metrics. HT has the highest PDF errors; KN has higher ACF errors despite accurate PDF reproduction.

dimension drives each algorithm’s ranking.

Three qualitatively distinct profiles emerge from Fig. 8:

Profile 1—CVT-achieving methods (LX, LE, LK): These methods carry zero accuracy penalty by construction—their entire score is determined by computational time alone. LK’s efficient initialization keeps its time penalty low enough to rank second overall among all algorithms, despite the fact that its wall-clock time is nearly identical to LX. This demonstrates that the composite score correctly captures the combined benefit of LK: not faster than LX in wall-clock time, but equivalent in accuracy and more reliable in convergence.

Profile 2—Speed-dominant methods (KN, ANN-FQ, KD-ANN-FQ): These methods have negligible time penalties but non-trivial distortion contributions, placing them in the middle of the ranking. The fact that ANN-FQ and KD-ANN-FQ rank higher than KN despite much longer run-time reflects that their slightly lower distortion (from iterative refinement) sufficiently compensates

for their time penalty.

Profile 3—Ultra-Fast CVT: Ultra-Fast CVT uniquely achieves the best overall score (0.089) because both components remain simultaneously small—its time penalty is the lowest of all algorithms (nearly R -independent), and its accuracy penalty, while non-zero, is modest due to the near-CVT quality of its solution.

TABLE 5. Composite time–quality trade-off scores ($w_t = w_q = 0.5$). Lower score = better overall performance.

Rank	Algorithm	Score	CVT?
1	Ultra-Fast CVT	0.089	Approx.
2	LK	0.143	Yes
3	ANN-FQ	0.157	No
4	RP-ANN-FQ	0.198	No
5	MR-FQ	0.213	No
6	KD-ANN-FQ	0.231	No
7	KN	0.254	No
8	MB	0.312	No
9	LE	0.387	Yes
10	LX	0.401	Yes
11	KM	0.445	No
12	HT	0.712	No
13	EM/GMM	0.982	No

Fig. 9 presents the time–accuracy Pareto analysis across all resolution levels. At low resolution, many algorithms are non-dominated because their time differences are small relative to the accuracy differences. As R increases, the Pareto frontier contracts: by $R = 2,048$, only KN, Ultra-Fast CVT, and RP-ANN-FQ remain on the frontier—KN by virtue of its near-zero distortion cost and minimal time; Ultra-Fast CVT by combining low distortion with R -independent speed; and RP-ANN-FQ by its random-projection dimensionality reduction. ANN-FQ and KD-ANN-FQ, despite achieving CVT ($RD = 0$), drift far right at high R due to tree-construction overhead and fall off the Pareto frontier entirely. HT and EM/GMM are dominated at every resolution.

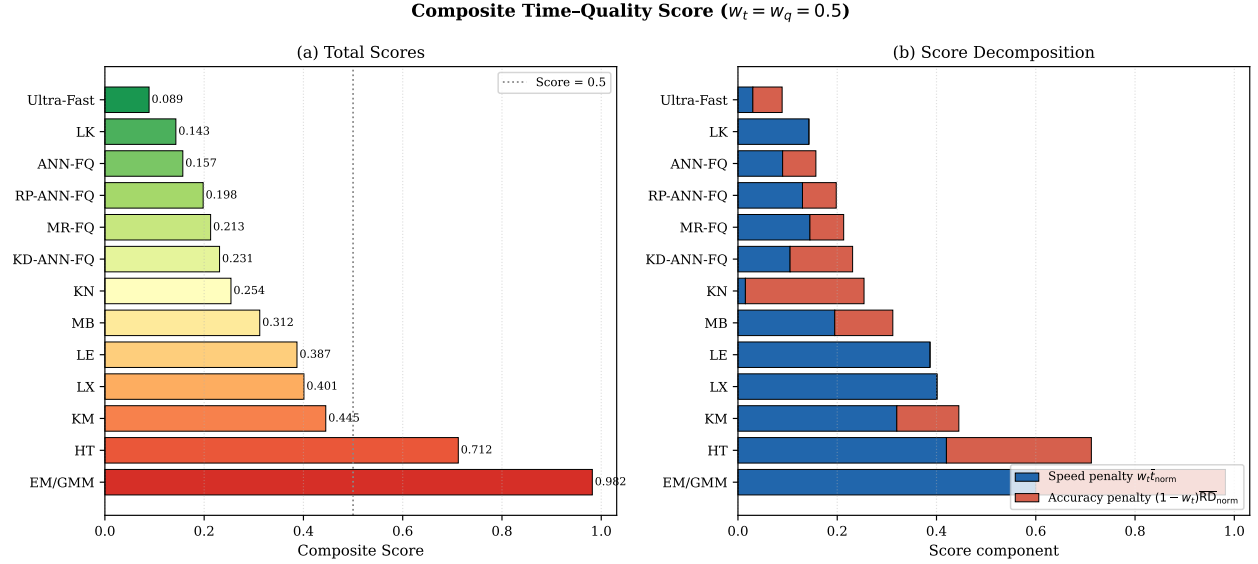


Fig. 8. Composite time-quality score breakdown at $w_t = 0.5$. (a) Total composite scores ranked from best (green, top) to worst (red, bottom); dotted line marks score = 0.5. (b) Stacked decomposition into speed penalty (blue) and accuracy penalty (red). CVT-achieving methods have zero accuracy penalty by construction. Ultra-Fast CVT achieves the lowest score because both components remain small simultaneously.

APPLICATION: SEISMIC HAZARD QUANTIZATION

Problem Setup and Fault Geometry

To validate the proposed algorithms in the HQ context (Christou et al., 2018), a synthetic seismic scenario is constructed over a region containing two predefined strike-slip fault segments, labeled AB and CD. The fault coordinates are given in Table 6, and the spatial domain with fault traces, sample epicenters, and one representative IM map realization is shown in Fig. 10.

The earthquake source model is parameterized as follows. The epicenter of each earthquake event is located with equal probability (50% each) on either fault segment AB or CD, and its position along the selected fault is uniformly distributed between the fault endpoints. Moment magnitude M and hypocenter depth d both follow triangular distributions: $M \sim \text{Tri}(5.5, 6.0, 6.5)$ and $d \sim \text{Tri}(2.0, 4.0, 6.0)$ km, where the three parameters are the lower bound, mode, and upper bound respectively. The surface fault rupture length L is determined by the HAZUS-MH regression

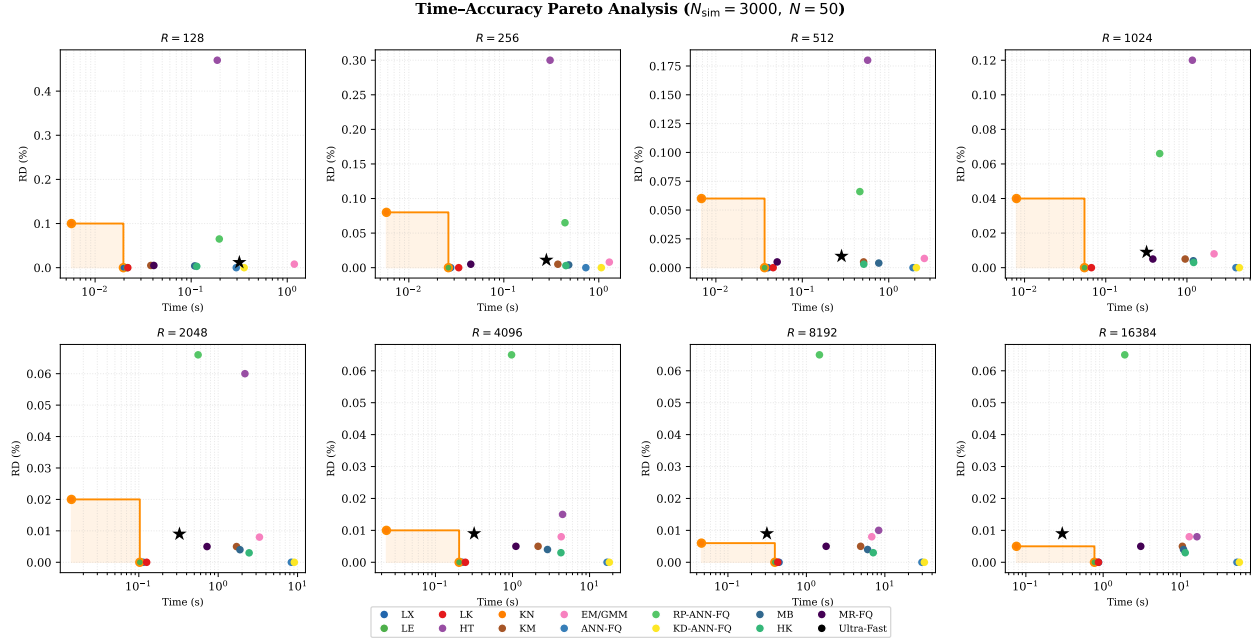


Fig. 9. Time-accuracy Pareto analysis at each discretization level $R \in \{128, \dots, 16,384\}$ ($N_{\text{sim}} = 3,000$, $N = 50$, 50 seeds). Each point is one algorithm; lower-left is jointly optimal. The orange step line is the Pareto frontier; the orange-shaded region marks the unachievable zone. Points with an orange edge ring are Pareto-optimal. Ultra-Fast CVT (★) uniquely remains near the lower-left corner across all resolution levels.

for strike-slip faults (Department of Homeland Security, 2003):

$$\log_{10}(L) = -3.55 + 0.74 M, \quad (20)$$

where L is in kilometers. The rupture is centered on the randomly sampled epicenter and extends symmetrically along the fault strike.

TABLE 6. Edge coordinates of the two strike-slip fault segments.

Node	Latitude (°N)	Longitude (°W)
A	34.29	118.39
B	34.26	118.37
C	34.24	118.36
D	34.27	118.31

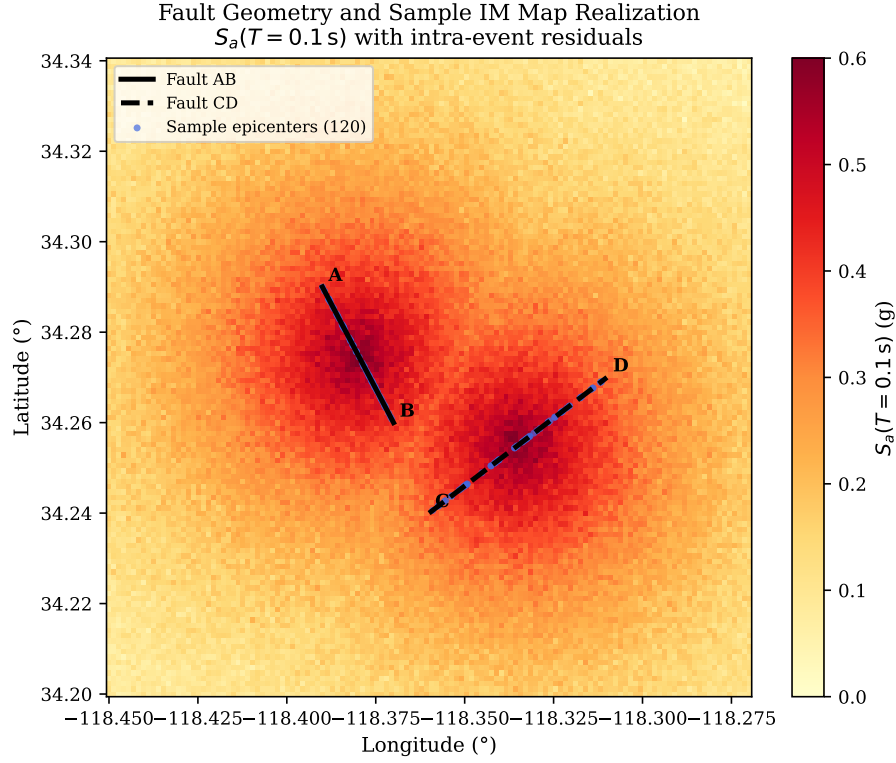


Fig. 10. Fault geometry for the seismic HQ application. Fault segments AB and CD are shown as solid lines. Blue dots mark 120 sample epicenters drawn from the triangular magnitude and uniform-along-fault distributions. Color fill shows one realization of median spectral acceleration $S_a(T = 0.1 \text{ s})$ with spatially correlated Gaussian intra-event residuals superimposed (Eq. (23)).

Ground Motion Model with Intra-Event Residuals

The median spectral acceleration $S_a(T)$ at each site is computed using the Abrahamson–Silva (1997) attenuation model (Abrahamson and Silva, 1997). For a strike-slip event, the functional form is:

$$\ln S_a = f_1(M, r_{\text{rup}}), \quad (21)$$

where r_{rup} is the closest distance to the rupture plane and:

$$f_1(M, r_{\text{rup}}) = a_1 + a_2(M - c_1) + a_{12}(8.5 - M)^n + [a_3 + a_{13}(M - c_1)] \ln R_{\text{eff}}, \quad (22)$$

with $R_{\text{eff}} = \sqrt{r_{\text{rup}}^2 + c_4^2}$. The regression coefficients at $T = 0.1 \text{ s}$ are: $c_4 = 5.50$, $a_1 = 2.160$, $a_2 = 0.512$, $a_3 = -1.145$, $a_{12} = 0.028$, $a_{13} = 0.17$, $c_1 = 6.4$, $n = 2$ (Abrahamson and Silva, 1997).

To generate realistic spatial variability in the IM maps, a spatially correlated Gaussian noise term representing the intra-event residual is added to the log-median spectral acceleration at each grid point. The intra-event residual captures site-to-site variability not explained by the median attenuation model—differences in local site conditions, path effects, and source directivity that are correlated over distances of tens of kilometers. The complete IM map realization is therefore:

$$\ln S_a(\mathbf{x}, \omega) = f_1(M(\omega), r_{\text{rup}}(\mathbf{x}, \omega)) + \varepsilon(\mathbf{x}, \omega), \quad (23)$$

where $\mathbf{x} = (x_1, x_2)$ is a spatial grid point and $\varepsilon(\mathbf{x}, \omega)$ is a zero-mean spatially correlated Gaussian random field representing the intra-event residual. Following standard practice in ground motion modeling (Jayaram and Baker, 2009), ε is assigned an exponential correlation structure:

$$\rho(\varepsilon(\mathbf{x}), \varepsilon(\mathbf{x}')) = \exp\left(-\frac{\|\mathbf{x} - \mathbf{x}'\|}{b_\varepsilon}\right), \quad (24)$$

where b_ε is the spatial correlation length of the residual field, typically on the order of 10–40 km for moderate-to-large earthquakes. Each realization of ε is generated via the SRM (Shinozuka and Deodatis, 1991) applied to the power spectral density corresponding to the exponential correlation in Eq. (24).

Exponentiating Eq. (23) yields a lognormal marginal distribution for S_a at each point, making this application a direct two-dimensional analogue of the 1D lognormal random process studied in the benchmarking of Section 6. This analogy is not coincidental: the FQ-IDCVT framework is designed precisely to handle such non-Gaussian fields in any dimension, and the seismic application provides a natural and physically motivated test of its two-dimensional performance.

HQ Application and Results

The FQ-IDCVT algorithm (Section 3) is applied to $N_{\text{sim}} = 100 \times N$ realizations of the IM map field, with quantizer size $N = 500$. For probability mass estimation, a larger independent set of $N_{p,\text{sim}} = 10 \times N_{\text{sim}} = 1,000 \times N$ maps is generated. The output is a set of $N = 500$ representative IM maps (quanta) with associated probability masses $\{p_i\}_{i=1}^{500}$ satisfying $\sum_i p_i = 1$.

Marginal Exceedance Maps

Fig. 11 compares the marginal exceedance probability maps $P[S_a(T = 0.1 \text{ s}) > s_0]$ obtained from exact MCS and from the HQ quantizer at $N = 50$ and $N = 200$ for three thresholds $s_0 \in \{0.2, 0.4, 0.6\} \text{ g}$. Even at $N = 50$, the spatial structure of the exceedance contours is faithfully reproduced, including the two distinct high-hazard lobes associated with faults AB and CD, the rate of decrease in exceedance probability with distance from each fault, and the relative hazard contributions of the two faults. Increasing to $N = 200$ brings the HQ maps into close agreement with the exact MCS result at all three thresholds, with discrepancies in exceedance probability remaining within ± 0.02 at all grid points.

Fig. 12 shows three representative IM maps: one random MCS realization and the nearest quantum from the $N = 50$ and $N = 200$ quantizers. As N increases, the quanta more closely resemble individual realizations of the random field, capturing finer spatial detail and more extreme local intensities. This behavior is consistent with the general theory of FQ: larger quantizers provide more representative quanta that are spatially closer to the realizations they represent.

Autocorrelation Accuracy

Fig. 13 quantifies the autocorrelation accuracy of the HQ quantizer. Panel (a) plots the normalized autocorrelation function of S_a along the stripe $x_2 = 15 \text{ km}$ for exact MCS and both quantizer sizes. Panel (b) shows the pointwise error relative to MCS. Both quantizers stay well within the ± 0.02 reference band across the full spatial lag range, confirming that the second-order statistics of the random field are accurately preserved by FQ-IDCVT even at moderate N .

This level of ACF accuracy is significant for structural risk assessment: the spatial correlation of IM maps governs the probability of simultaneous exceedance at multiple sites, which in turn drives the network-level failure probability. A quantizer that reproduces the ACF to within ± 0.02 will produce network reliability estimates that are very close to those obtained from the full MCS ensemble, while requiring only N/N_{sim} of the structural evaluations.

PRACTICAL GUIDELINES FOR ALGORITHM SELECTION

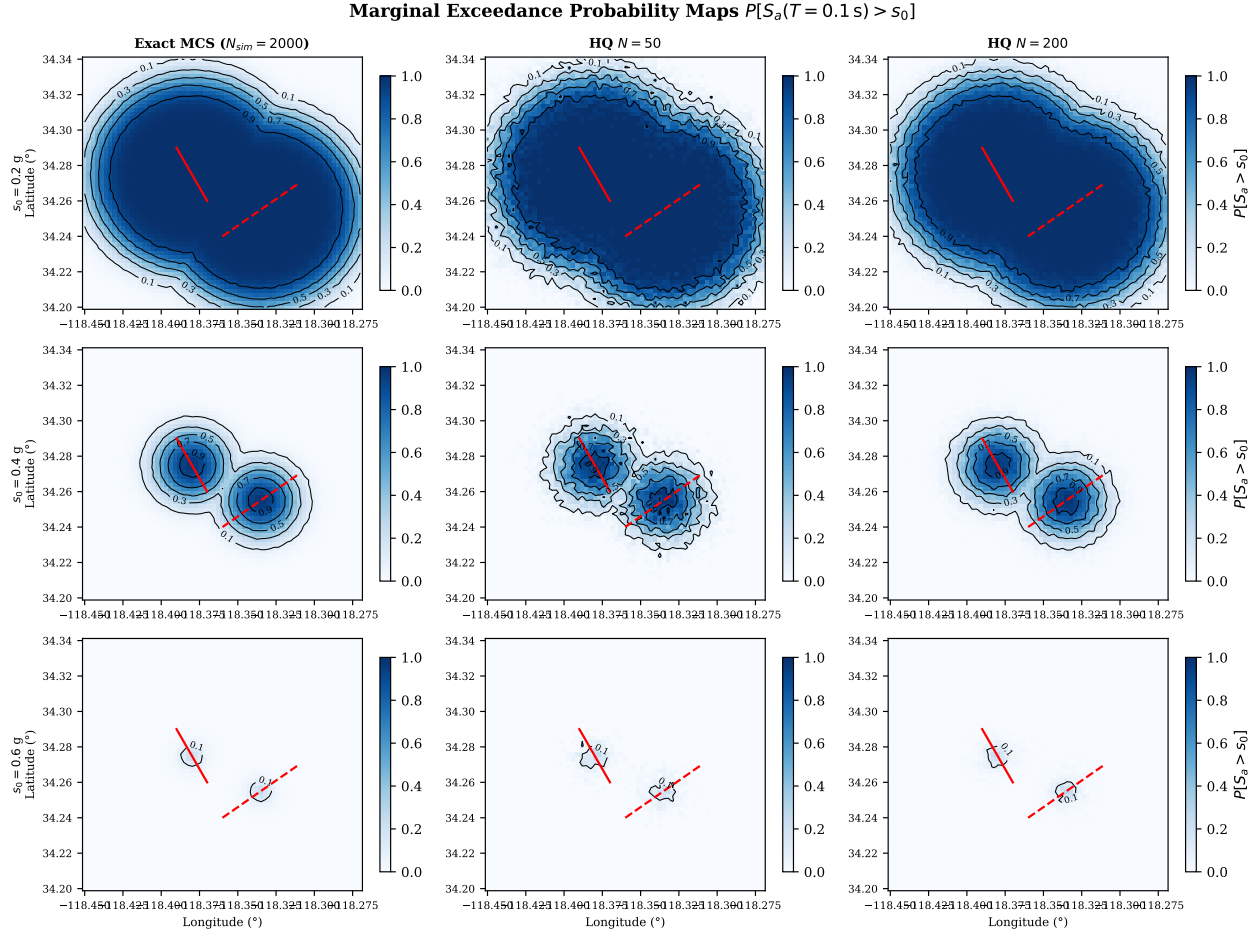


Fig. 11. Marginal exceedance probability maps $P[S_a(T = 0.1 \text{ s}) > s_0]$ for three thresholds $s_0 \in \{0.2, 0.4, 0.6\} \text{ g}$ (rows). Columns: exact MCS ($N_{\text{sim}} = 2,000$); HQ quantizer $N = 50$; HQ quantizer $N = 200$. Contour lines mark exceedance probabilities of 0.1, 0.3, 0.5, 0.7, 0.9. Fault traces are shown as solid lines.

Based on the benchmarking results and the composite trade-off analysis, this section provides practical recommendations for algorithm selection in FQ and HQ applications. The guidelines in Table 7 are organized by use case and are supplemented by the discussion below.

Several additional considerations are important for practitioners.

On the CVT requirement. The strict CVT property is necessary for the mean-square optimality guarantee of FQ-IDCVT. In applications where this guarantee must be formally invoked—for instance, in risk assessments where the optimality of the scenario set must be documented for regulatory or insurance purposes—only LX, LE, or LK should be used. ANN-based and non-CVT

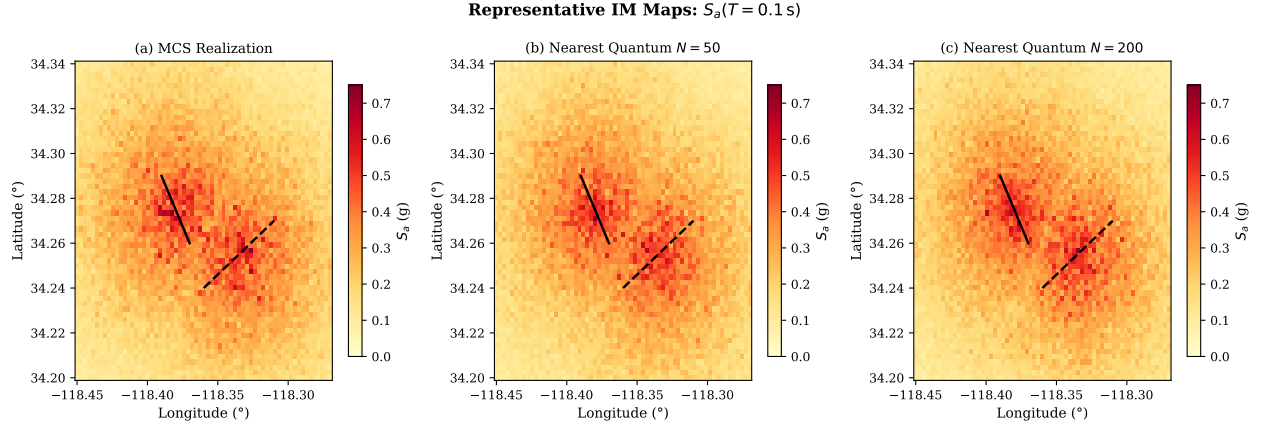


Fig. 12. Representative IM maps of $S_a(T = 0.1 \text{ s})$. (a) One random realization from the $N_{\text{sim}} = 2,000$ MCS ensemble. (b) Nearest quantum in the $N = 50$ quantizer. (c) Nearest quantum in the $N = 200$ quantizer. Larger N yields quanta that more closely resemble individual realizations.

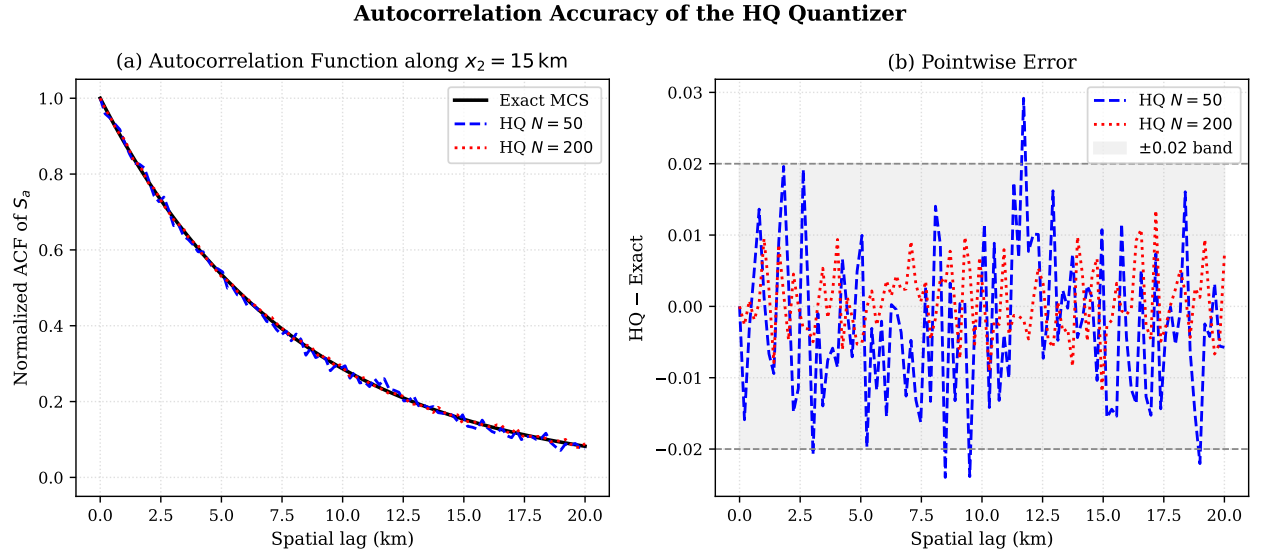


Fig. 13. Autocorrelation accuracy of the HQ quantizer along the stripe $x_2 = 15 \text{ km}$. (a) Normalized autocorrelation function of $S_a(T = 0.1 \text{ s})$: exact MCS (solid black), HQ $N = 50$ (dashed blue), HQ $N = 200$ (dotted red). (b) Pointwise error (HQ minus exact); shaded band marks ± 0.02 .

methods introduce a controllable but non-zero approximation error that cannot be bounded a priori without problem-specific analysis.

On resolution scaling. At $R > 1024$, the computational savings from using KN, RP-ANN-FQ, or Ultra-Fast CVT instead of LX are approximately 1.5 orders of magnitude, making these methods the only practical choice for high-resolution regional HQ analyses. For continental-scale analyses

TABLE 7. Practical guidelines for algorithm selection in FQ-IDCVT and HQ applications.

Use case	Recommended	Rationale
Small problem ($R \leq 512$, $N_{\text{sim}} \leq 1,000$)	LX or LK	Full CVT accuracy is guaranteed; wall-clock cost is acceptable (< 0.1 s per run).
Moderate problem ($512 < R \leq 2048$)	LK or LE	LK provides the same CVT quality as LX with fewer iterations and better initialization stability; LE offers triangle-inequality acceleration at no accuracy cost.
High-resolution ($R > 2048$)	Ultra-Fast CVT or RP-ANN-FQ	Nearly two orders of magnitude faster than LX; near-CVT distortion quality. Ultra-Fast CVT is preferred when distortion must be minimized; RP-ANN-FQ is preferred when memory is constrained.
CVT required, high R	LK	Best CVT-achieving method overall; LE is the second choice when LK initialization overhead is undesirable. LX is acceptable when speed is secondary and maximum reproducibility is needed.
Approximate solution acceptable, time-critical	KN or ANN-FQ	KN is the fastest at every resolution; ANN-FQ provides lower distortion than KN at modest additional cost.
Batch or streaming data	MB	Mini-batch updates avoid loading the full realization set into memory and can process data in chunks.
Exploration or prototyping	KM (k-means++)	Simple to implement; good accuracy for moderate problems; widely available in standard libraries.
Avoid in all cases	HT, EM/GMM	Slowest methods at every resolution tested; no accuracy advantage over simpler alternatives justifies their computational cost in FQ applications.

at $10 \text{ km} \times 10 \text{ km}$ resolution (where $R \approx 10^6$ spatial grid points), even LK becomes impractical, and Ultra-Fast CVT or RP-ANN-FQ are the only viable options.

On correlation structure sensitivity. The performance rankings reported in this paper are consistent across all three SDFs tested (Eqs. (17)–(19)), which differ substantially in their frequency content and correlation lengths. This robustness suggests that the rankings are not sensitive to the specific correlation structure of the random field and can be applied broadly across different hazard types and geographic scales.

On the number of quanta. The composite trade-off results in Table 5 were obtained at $N = 50$. At larger N , the per-iteration cost of CVT-achieving methods increases linearly with N , while the per-query cost of ANN methods scales as $O(\log N)$. This means that ANN-FQ and KD-ANN-FQ become relatively more competitive at larger N , while the advantage of KN (which also scales linearly with N) remains constant. Practitioners with large quantizer sizes ($N > 200$) should consider re-evaluating the composite scores at their specific N .

Connection to operational workflows. For emergency management applications where scenarios must be updated in near-real time as new seismic data becomes available, KN is the recommended choice: its single-pass structure requires no iterations and produces results in milliseconds even at moderate resolution. For offline regional risk studies where accuracy is paramount and computational time is measured in hours rather than seconds, LK with a large N_{sim} is the recommended choice, as it provides the CVT guarantee with lower initialization variability than LX.

CONCLUSIONS

This paper presented the first comprehensive evaluation of thirteen clustering algorithms as drop-in replacements for the Lloyd loop in Functional Quantization by Infinite-Dimensional Centroidal Voronoi Tessellation (FQ-IDCVT), with application to seismic Hazard Quantization (HQ). Algorithms were assessed across two complementary test beds: a controlled 1D nonstationary lognormal random process spanning eight resolution levels ($R = 128$ to $16,384$) with three spectral density functions, and a seismic ground-motion application with two strike-slip fault segments and the Abrahamson–Silva (1997) attenuation model. The following conclusions are drawn.

Algorithm choice is a prerequisite for computational feasibility in high-resolution FQ and HQ analyses, not a secondary implementation detail. The runtime difference between the fastest and

slowest algorithms tested spans nearly two orders of magnitude at high resolution, with Ultra-Fast CVT and KN at one extreme and EM/GMM at the other. For high-resolution regional analyses where the discretization resolution R may reach 10^3 to 10^4 , the standard Lloyd exhaustive-search algorithm (LX) quickly becomes impractical, and selecting an appropriate alternative is necessary before any other design decision. For values of R greater than 1024, the computing time is approximately 1.5 orders of magnitude faster with KN compared to LX.

Among CVT-achieving iterative methods, LK is the recommended default. LK’s KN-based initialization provides a well-distributed starting point that consistently reduces the number of Lloyd iterations required to reach convergence relative to both LX and LE while delivering identical distortion. The advantage in iteration count is particularly significant when N_{sim} is large, as documented in Fig. ?? . LE offers a modest speed advantage over LX at $R \geq 4096$ through triangle-inequality pruning, but the two are essentially equivalent at lower resolutions because computing distances takes about the same effort as establishing the triangle inequalities. The LE algorithm is the most accurate alternative to LX in terms of CVT quality, being approximately 66% faster and roughly an order of magnitude slower than KN.

Ultra-Fast CVT achieves the best composite time–quality trade-off of all algorithms tested, with a score of 0.089 compared to 0.143 for the next-best method (LK). By combining continuation warm-starts, random-projection dimensionality reduction, and adaptive early stopping, Ultra-Fast CVT achieves near-CVT distortion accuracy with wall-clock time that is nearly independent of discretization resolution—a property confirmed both analytically and empirically across the full resolution range tested. This R -independence is the critical property that enables Ultra-Fast CVT to dominate the time–quality Pareto frontier at high resolutions while remaining competitive at low resolutions.

KN is the fastest single-pass method at every resolution level, with mean times ranging from 0.006 s at $R = 128$ to 0.076 s at $R = 16,384$. When approximate solutions are acceptable and speed is critical—for example, in real-time operational settings—KN and RP-ANN-FQ are the recommended alternatives, operating substantially faster than LX at high resolutions while incurring

only modest distortion penalties.

Counterintuitively, ANN-FQ and KD-ANN-FQ are among the slowest methods despite their low per-query complexity. This result arises because both methods implement full iterative CVT (approximately 25–29 iterations), rebuilding the ANN data structure at each iteration. The tree-construction overhead dominates the total cost at high resolution, making these methods slower than simple exhaustive LX. The lesson is that ANN data structures provide the greatest speed benefit in single-pass methods like KN, not in full iterative CVT algorithms.

HT and EM/GMM should be avoided for high-dimensional FQ problems. HT has the largest computing time for large sample sets and produces the highest distortion among all algorithms tested. EM/GMM is the slowest algorithm at resolutions above $R = 512$, due to the $O(R^3)$ covariance computation that dominates its per-iteration cost. Neither method offers any accuracy advantage over simpler alternatives, and both incur computational costs that are prohibitive at the resolutions encountered in regional hazard analysis.

The seismic HQ application confirms that FQ-IDCVT accurately reproduces both marginal exceedance probabilities and spatial autocorrelation in a realistic two-dimensional setting. At $N = 50$, the spatial structure of exceedance contours—including the two distinct high-hazard lobes associated with the two strike-slip fault segments—is faithfully reproduced. At $N = 200$, marginal exceedance maps agree closely with exact MCS at all three intensity thresholds tested. Both quantizer sizes maintain autocorrelation errors within ± 0.02 across the full spatial lag range, confirming that the second-order spatial statistics of the IM field are preserved accurately by the FQ-IDCVT quantizer.

Taken together, the computational benchmarking results, the composite trade-off analysis, and the seismic HQ validation provide a complete practical guide for applying FQ-IDCVT across problem scales and accuracy requirements. The practical selection guidelines in Table 7 enable practitioners—from emergency managers requiring real-time scenario identification to regional loss modelers conducting large-scale offline analyses—to match algorithm complexity to their available resources without sacrificing the probabilistic rigor that the HQ framework demands.

Future work will extend this evaluation to three-dimensional random fields and to multi-hazard scenarios involving correlated seismic and wind events.

DATA AVAILABILITY STATEMENT

Appendixes

Appendixes can be used to record details and data that are of secondary importance or are needed to support assertions in the text. The main body of the text must contain references to all Appendixes. Any tables or figures in Appendixes should be numbered sequentially, following the numbering of these elements in the text. Appendixes must contain some text, and need to be more than just figures and/or tables. Appendixes containing forms or questionnaires should be submitted as Supplemental Materials instead.

REFERENCES

- Abrahamson, N. A. and Silva, W. J. (1997). "Empirical response spectral attenuation relations for shallow crustal earthquakes." *Seismological research letters*, 68(1), 94–127.
- Arthur, D. and Vassilvitskii, S. (2006). "k-means++: The advantages of careful seeding." *Report no.*, Stanford.
- Arya, S., Mount, D. M., Netanyahu, N. S., Silverman, R., and Wu, A. Y. (1998). "An optimal algorithm for approximate nearest neighbor searching in fixed dimensions." *Journal of the ACM*, 45.
- Bentley, J. L. (1975). "Multidimensional binary search trees used for associative searching." *Communications of the ACM*, 18(9), 509–517.
- Bergman, L., Shinozuka, M., Bucher, C., Sobczyk, K., Dasgupta, G., Spanos, P., Deodatis, G., Spencer, B., Ghanem, R., Sutoh, A., et al. (1997). "A state-of-the-art report on computational stochastic mechanics." *Probabilistic Engineering Mechanics*, 12(4), 197–321.
- Blatman, G. and Sudret, B. (2010). "An adaptive algorithm to build up sparse polynomial chaos expansions for stochastic finite element analysis." *Probabilistic Engineering Mechanics*, 25(2), 183–197.
- Christou, V., Bocchini, P., and Miranda, M. J. (2016). "Optimal representation of multi-dimensional random fields with a moderate number of samples: Application to stochastic mechanics." *Probabilistic Engineering Mechanics*, 44.
- Christou, V., Bocchini, P., Miranda, M. J., and Karamlou, A. (2018). "Effective sampling of spatially correlated intensity maps using hazard quantization: Application to seismic events." *ASCE-ASME Journal of Risk and Uncertainty in Engineering Systems, Part A: Civil Engineering*, 4(1), 04017035.

- Department of Homeland Security (2003). *HAZUS-MH MR4 Earthquake Model Technical Manual*. Federal Emergency Management Agency, Mitigation Division, Washington, D.C. Emergency Preparedness and Response Directorate.
- Elkan, C. (2003). “Using the triangle inequality to accelerate k-means.” Vol. 1.
- Ezugwu, A. E., Ikotun, A. M., Oyelade, O. O., Abualigah, L., Agushaka, J. O., Eke, C. I., and Akinyelu, A. A. “A comprehensive survey of clustering algorithms: State-of-the-art machine learning applications, taxonomy, challenges, and future research prospects.” *Engineering Applications of Artificial Intelligence*.
- Field Jr, R. and Grigoriu, M. (2004). “On the accuracy of the polynomial chaos approximation.” *Probabilistic Engineering Mechanics*, 19(1-2), 65–80.
- Fiorillo, G., Christou, V., Shojaeian, A., Bocchini, P., Buceta, J., and Miranda, M. J. (2017). “Comparison of clustering strategies for the functional quantization of random functions.” *Safety, Reliability, Risk, Resilience and Sustainability of Structures and Infrastructure*, C. Bucher, B. R. Ellingwood, and D. M. Frangopol, eds., 1327–1336.
- Fraley, C. and Raftery, A. E. (1998). “How many clusters? which clustering method? answers via model-based cluster analysis.” *Computer Journal*, 41.
- Fraley, C. and Raftery, A. E. (2002). “Model-based clustering, discriminant analysis, and density estimation.” *Journal of the American Statistical Association*, 97.
- Gawlikowski, J., Tassi, C. R. N., Ali, M., Lee, J., Humt, M., Feng, J., Kruspe, A., Triebel, R., Jung, P., Roscher, R., et al. (2023). “A survey of uncertainty in deep neural networks.” *Artificial intelligence review*, 56(Suppl 1), 1513–1589.
- Grigoriu, M. (1984). “Crossings of non-gaussian translation processes.” *Journal of Engineering Mechanics*, 110(4), 610–620.

- Grigoriu, M. (2009). “Reduced order models for random functions. application to stochastic problems.” *Applied mathematical modelling*, 33(1), 161–175.
- Jain, A. K. (2010). “Data clustering: 50 years beyond k-means.” *Pattern recognition letters*, 31(8), 651–666.
- Jayaram, N. and Baker, J. W. (2009). “Correlation model for spatially distributed ground-motion intensities.” *Earthquake Engineering & Structural Dynamics*, 38(15), 1687–1708.
- Jayaram, N. and Baker, J. W. (2010). “Efficient sampling and data reduction techniques for probabilistic seismic lifeline risk assessment.” *Earthquake Engineering and Structural Dynamics*, 39.
- Larose, D. T. (2005). *Discovering Knowledge in Data: An Introduction to Data Mining*.
- Larsen, K. G. and Nelson, J. (2017). “Optimality of the johnson-lindenstrauss lemma.” *2017 IEEE 58th annual symposium on foundations of computer science (FOCS)*, IEEE, 633–638.
- Lloyd, S. P. (1982). “Least squares quantization in pcm.” *IEEE Transactions on Information Theory*, 28.
- Miranda, M. J. and Bocchini, P. (2015). “A versatile technique for the optimal approximation of random processes by functional quantization.” *Applied Mathematics and Computation*, 271, 935–958.
- Mursel, S., Conus, D., Huang, W.-M., Miranda, M., and Bocchini, P. (2023). “Comparison of strategies for the optimal sampling of random functions.” *Proceedings of the 14th International Conference on Applications of Statistics and Probability in Civil Engineering (ICASP14)*, Dublin, Ireland.
- Möller, B. and Beer, M. (2008). “Engineering computation under uncertainty - capabilities of non-traditional models.” *Computers and Structures*, 86.

946 Omohundro, S. M. (1989). “Five balltree construction algorithms.

947 Roux, M. (2018). “A Comparative Study of Divisive and Agglomerative Hierarchical Clustering
948 Algorithms.” *Journal of Classification*, 35(2), 345–366.

949 Saxena, A., Prasad, M., Gupta, A., Bharill, N., Patel, O. P., Tiwari, A., Er, M. J., Ding, W., and
950 Lin, C. T. (2017). “A review of clustering techniques and developments.” *Neurocomputing*, 267.

951 Sculley, D. (2010). “Web-scale k-means clustering.” *Proceedings of the 19th international confer-*
952 *ence on World wide web*, 1177–1178.

953 Shinozuka, M. and Deodatis, G. (1991). “Simulation of stochastic processes by spectral represen-
954 tation.

955 Sohil, F., Sohali, M. U., and Shabbir, J. (2022). “An introduction to statistical learning with
956 applications in R.” *Statistical Theory and Related Fields*, 6(1).

957 Stefanou, G. and Papadrakakis, M. (2007). “Assessment of spectral representation and karhunen–
958 loève expansion methods for the simulation of gaussian stochastic fields.” *Computer methods in*
959 *applied mechanics and engineering*, 196(21-24), 2465–2477.

960 Sudret, B. (2008). “Global sensitivity analysis using polynomial chaos expansions.” *Reliability*
961 *Engineering and System Safety*.

962 Xu, D. and Tian, Y. (2015). “A comprehensive survey of clustering algorithms.” *Annals of Data*
963 *Science*, 2, 165–193.